Final Report

Biomass-to-Jet: Using Gasification and Fischer–Tropsch Chemistry

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Executive Summary

This report evaluates the technical, economic, and environmental feasibility of producing Sustainable Aviation Fuel (SAF) using a biomass-to-liquids process based on gasification and Fischer–Tropsch (FT) synthesis. The proposed facility is designed to convert approximately 5,700 tonnes/day of woody biomass into synthetic paraffinic kerosene (SPK), with a target production capacity of roughly 10,000 barrels per day. The plant is located in St. James Parish, Louisiana, leveraging strong regional biomass availability, existing infrastructure, and proximity to Gulf Coast refining and hydrogen networks.

The process consists of four major sections: biomass gasification and syngas cleanup, FT synthesis, hydrotreating, and hydrocracking. This configuration enables the conversion of low-value biomass into high-quality, low-sulfur jet fuel that meets ASTM blending specifications. Due to the olefinic nature of the produced fuel, the SAF is blended at a 1:1 ratio with conventional jet fuel to meet final fuel standards.

From an economic standpoint, the total capital investment for the plant is estimated at approximately $980 million, with annual operating costs of $288 million and total revenue of $747 million, including fuel credits. The project demonstrates strong profitability, with a discounted cash flow rate of return (DCFROR) of approximately 32% and a payback period of just over two years. However, the analysis also shows that profitability is highly dependent on policy incentives; without fuel credits, the project becomes economically unfavorable.

A sensitivity analysis indicates that fuel price (revenue) is the most significant factor affecting profitability, followed by capital cost and variable operating costs. Fixed costs and

depreciation methods have minimal impact. These results highlight the importance of market conditions and cost control in determining project success.

Environmental analysis shows that the primary emissions are carbon dioxide and non-hazardous flue gases, while liquid and solid wastes are manageable with standard treatment methods. The process offers meaningful reductions in lifecycle emissions compared to conventional jet fuel, particularly when paired with green hydrogen and renewable energy sources.

Based on these findings, the proposed process is technically viable and economically attractive under current incentive structures. Final recommendations include reducing raw material costs through onsite production of renewable hydrogen and oxygen via electrolysis, improving process efficiency through heat integration and catalyst optimization, and exploring carbon capture technologies to further enhance sustainability and long-term competitiveness.

1. Introduction

1.1 Problem Statement

Aviation is one of the most energy-intensive transportation modes and is inherently difficult to decarbonize due to strict safety regulations, long asset lifetimes, and the absence of cost-effective alternatives to high-density liquid fuels. Conventional jet fuel is derived from fossil crude oil through distillation and hydrotreating, and its combustion contributes significantly to global greenhouse gas emissions, as well as particulate matter and NOx, which impact both climate and air quality. Worldwide, aviation accounts for 2% of all carbon dioxide (CO_2) emissions and 12% of all CO_2 emissions from transportation. Global and national policy initiatives are establishing targets to decarbonize the aviation sector. ICAO's CORSIA program

caps net CO_2 emissions from international aviation at 2020 levels through 2035, and the aviation industry has set a goal of reaching net-zero emissions by 2050. Because electrification and hydrogen are unlikely to be widely deployed in the near term, Sustainable Aviation Fuel (SAF) represents the most practical pathway to meet these objectives. In the U.S., the 2021 Sustainable Aviation Fuel Grand Challenge aims to scale domestic SAF production to 3 billion gallons annually by 2030 and 35 billion gallons annually by 2050, with a lifecycle reduction in GHG emissions of at least 50%, thereby creating long-term demand and policy support for technologies such as FT-SPK.

Sustainable Aviation Fuel (SAF) is an alternative fuel made from non-petroleum feedstocks such as biomass, waste oils, agricultural residues, and municipal solid waste. SAF has emerged as the most viable pathway for reducing aviation emissions because it is a drop-in fuel that can meet the same performance standards as Jet A without requiring modifications to jet engines, fuel systems, or airports. This is critical for near-term decarbonization because the physical and regulatory constraints of aviation make it impractical to implement rapid changes in propulsion systems, fuel storage, and aircraft design. SAF must therefore meet stringent performance specifications, including energy density, combustion characteristics, volatility, and cold-flow properties, while offering substantial lifecycle reductions in greenhouse gas emissions.

Multiple production pathways for SAF have been developed and approved under ASTM International standards, including Hydro processed Esters and Fatty Acids (HEFA), Alcohol-to-Jet (ATJ), and Fischer-Tropsch Synthetic Paraffinic Kerosene (FT-SPK). Among these, FT-SPK has gained significant attention due to its ability to convert a diverse range of low-value, sustainable feedstocks into high-quality paraffinic hydrocarbons with ultralow sulfur and aromatic content. When produced from biomass or waste-derived syngas, FT-SPK can provide

substantial lifecycle emission reductions and additionally support waste diversion efforts, contributing both environmental and societal benefits.

Despite these advantages, SAF production remains limited by high capital intensity, feedstock logistics, and operational complexity, resulting in production volumes that comprise less than 1% of global jet fuel demand. However, the combination of international emissions mandates, national policy incentives, and long-term corporate sustainability commitments is driving rapid development of the SAF sector. In this context, evaluating the technical, economic, and environmental performance of FT-SPK production is essential to determine optimal process configurations, assess scalability, and identify barriers to commercialization. This project focuses on the design and preliminary analysis of a commercial-scale FT-SPK process capable of supporting the emerging demand for low-carbon aviation fuels.

1.2 Market Analysis

The Sustainable Aviation Fuel (SAF) market is poised for significant growth, driven by increasing environmental concerns, regulatory pressures, and advancements in fuel production technologies. Despite its promising potential, SAF production faces challenges, including high capital costs for establishing production facilities, difficulties in securing consistent feedstocks, and operational complexities. These factors contribute to the high production cost of SAF, which currently ranges from $3.50 to $6.00 per gallon, significantly higher than traditional jet fuel, which typically costs between $1.50 and $2.00 per gallon. This price disparity presents a barrier to widespread adoption. However, as production scales up, technology improves, and government incentives such as tax credits, subsidies, and other financial support are implemented, the price of SAF is expected to decrease. For example, the U.S. SAF Grand

Challenge aims to produce 3 billion gallons of SAF annually by 2030, which will help lower production costs through economies of scale and infrastructure development.

The global SAF market is projected to grow from USD 2.7 billion in 2025 to USD 28.6 billion by 2032, at a CAGR of 39.95%. North America is expected to remain the largest market, accounting for 46% of the market share by 2024, largely due to strong government-backed initiatives such as the SAF Grand Challenge. Europe is also a key player, driven by the European Green Deal and national goals for carbon-neutral aviation. Meanwhile, the Asia-Pacific region is set to experience rapid growth, fueled by rising investments, increasing demand for cleaner fuels, and technological advancements in SAF production. The combination of these factors, alongside the scaling of SAF production, is expected to drive down costs and make SAF more competitive with traditional jet fuel, supporting its long-term adoption in the aviation industry.

1.3 Judging Criteria

The Fischer-Tropsch (FT) pathway was selected over alternative sustainable aviation fuel production pathways, based on its superior feedstock flexibility, blending capability, scalability, long-term feasibility for meeting the growing SAF demand, and its potential for reducing carbon emissions during the life cycle of the product.

One major advantage of FT is its ability to process a broad range of carbon-based feedstocks like woody biomass, agricultural, Municipal Solid Waste (MSW), etc. The design basis of the plant accounts for this advantage by choosing St. James Parish, Louisiana, where forestry residue is abundant at a low cost. In contrast, other competing processes like HEFA are constrained almost exclusively to rich lipid feedstocks which are highly limited. These feedstock

sources are heavily competed for by the renewable diesel market, leaving little to no room for the SAF market. Alcohol to Jet (ATJ) feedstocks depend on sugar, starch, or cellulosic ethanol. Food and agricultural waste have a highly competitive market as these feeds are often used in biofuel industries to produce things like biodiesel. Because FT can utilize non-food, non-lipid, waste biomass at very large scales, it provides a more robust and sustainable feedstock supply chain.

Fisher-Tropsch derived synthetic paraffinic kerosene (FT-SPK), also known as jet fuel, is certified under the ASTM D7566 Annex A1 to be blended at up to 50 vol% with conventional Jet A/A-1 jet fuel. This high blending limit provides two critical advantages. It enables significant displacement of fossil fuel derived jet fuel per unit of SAF produced and allows for drop-in use without any modifications to aircraft engines, fuel handling systems, or airport logistics that are designed to operate with fossil fuel derived kerosene. Other pathways offer a lower maximum blend limit ranging from 10% to 30% thus FT offers both high blending capabilities and feedstock scalability.

The FT pathway also enables deep decarbonization because the biomass to liquids routes achieves significant carbon intensity reductions depending on feedstock logistics and hydrogen sourcing. Other pathways simply do not offer this level of decarbonization due to FT's ability to reduce greenhouse gas emissions.

The long-term expansion potential for FT-SPK is very promising as SAF initiatives and demands continue to increase. Existing commercial plants demonstrate that 10,000 barrels/day is sustainable and efficient making the design of this plant strategic. Competing processes are generally smaller in scale, are limited by supply of feedstocks, and are less aligned with long-term SAF volume requirements mandated by U.S and EU policy. The U.S. Gulf Coast also

already supports biomass supply chains, gasification and syngas-handling experience, hydrogen production and contains existing refinery and upgrading systems.

In summary, the Fisher-Tropsch pathway was selected because it provides broad and abundant feedstock options, 50% blend certification, which matches or exceeds alternatives, superior scalability for large volume SAF production, and lower carbon intensity. For these reasons, FT to SAF offers the most robust, sustainable, and scalable pathway for meeting long-term aviation decarbonization targets compared to methods like HEFA or ATJ.

2. Process Description

2.1 Chosen Process

The general Fischer-Tropsch process to produce synthetic paraffinic kerosene, also known as Jet Fuel, is as follows. The first step is feedstock selection, which can be any carbon-based material known as biomass. In this case, wood-based biomass is gasified to create synthetic gas, syngas, which is primarily a mixture of carbon monoxide and hydrogen gas ($CO_{(g)} + H_{2\,(g)}$). The syngas must then undergo cleaning to ensure that a relatively low concentration of carbon dioxide (CO_2) remains. The lean syngas then undergoes catalytic reactions in an FT reactor that uses cobalt as a catalyst in low temperatures to form long chain hydrocarbons. These hydrocarbons are then sent to upgrading and fractionation processes to produce synthetic paraffinic kerosene (SPK) that meets the ASTM D7566 standard. The final product once blended at up to 50% must meet the new ASTM D1655 standard.

The detailed process will follow the major sections of the block flow diagram highlighted below in which include biomass gasification and syngas cleanup, Fischer-Tropsch reactions,

hydrotreating, and hydrocracking. The operating conditions are based on the process simulation modeled using ASPEN Plus v14

The process begins with the gasification of wood-based biomass to produce syngas which is then cleaned to produce lean syngas that are fed into the FT reactor. For this process, the necessary feed is found in the design basis section to be approximately 5,700 tonnes/day of biomass. Unlike fossil-based syngas production, biomass gasification produces syngas containing significant amounts of CO_2, particulates and ash, as well as other contaminants. Extensive gas cleanup and conditioning are required prior to FT synthesis.

The first step in the biomass gasification process, shown in **Error! Reference source not found.**, is to heat and dry the incoming biomass in a dryer (*D-100*) operating at about 250°F to reduce the typical inlet moisture of 30-50 wt.% down to <10 wt.%. Air is fed into the dryer after being heated and it expels the dry biomass and excess moisture. The feed is then ready to be sent to the biomass gasifier (*R-100*) where the dried biomass along with oxygen is converted into raw syngas via partial oxidation reactions with oxygen. The primary reactions are shown below (Equation 1).

$$C + O_2 \rightarrow CO_2$$

$$C + {}^{1}/_{2}\, O_2 \rightarrow CO$$

$$C + H_2O \rightarrow CO + H_2$$

$$CO + H_2O \leftrightarrow CO_2 + H_2$$

Equation 1: *Primary Biomass Conversion Equations*

Assuming a fluidized bed reactor is used then the temperature can be specified at ~1400°F with pressure an operating pressure of ~15 psia. The products of this reactor include raw syngas and tar.

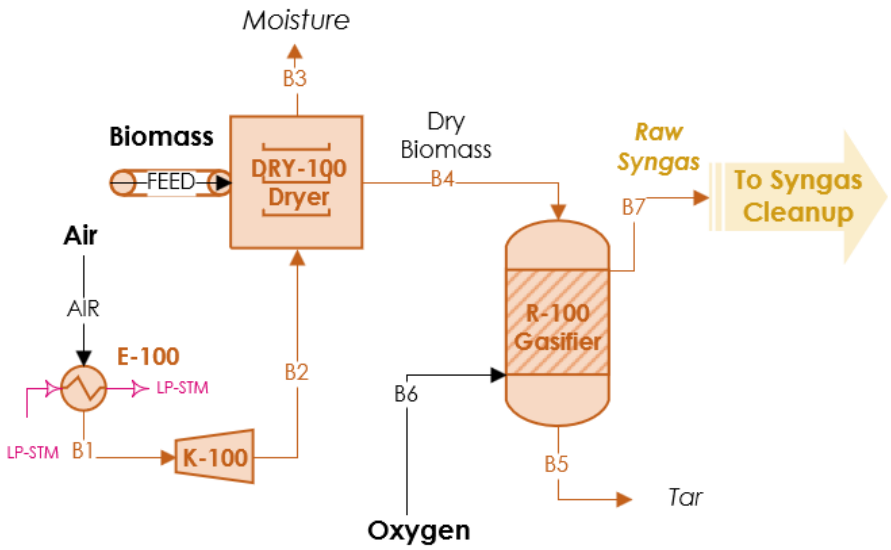


Figure 1: *Biomass Gasification Process Flow Diagram*

The cleanup process is as show in ***Error! Reference source not found.*** below. Raw Syngas is fed into the cyclone *C-100* removes fine ash and char particles from raw syngas to protect the downstream equipment and operates at ~1400°F. The incoming gas will be compressed to a new pressure of 180 psia. E-100 then cools the hot syngas to enable downstream wet scrubbing and contaminant removal. Following cooling, the syngas scrubber *V-100* removes remaining contaminants using water and operates at 100°F and ~180pisa. The knockout drum *V-101* removes entrained liquid droplet and condensed liquids from scrubbed syngas at 86°F. The

water-gas shift reactor *R-101* adjusts the H_2/CO ratio to approximately 1.7, as determined in the design basis and required for cobalt FT synthesis. The main reaction here is shown below (Equation 2). The reactor is expected to operate at high temperatures ~212°F) and at approximately atmospheric pressure 14.5 psia. From the design basis the steam to dry syngas ratio can be taken to be 0.5kg/kg.

$$CO + H_2O \leftrightarrow CO_2 + H_2$$

Equation 2: *Water-Gas Shift Reaction*

The final *V-102* absorber reduces the concentration of CO_2 to improve FT reactor productivity and operates at 145 psia removing 70-90% of CO_2. The resulting lean, conditioned syngas meets the FT feed specifications of having a H_2/CO ratio of 1.7, particulate concentration < 0.1 mg/Nm^3, and negligible concentration of sulfur and nitrogen species.

Key assumptions include that woody residues will have a consistent composition, air-blown gasification is acceptable, heat integration will greatly reduce external utility demands, and overall biomass to syngas efficiency aligns with reported ~70% cold gas efficiency.

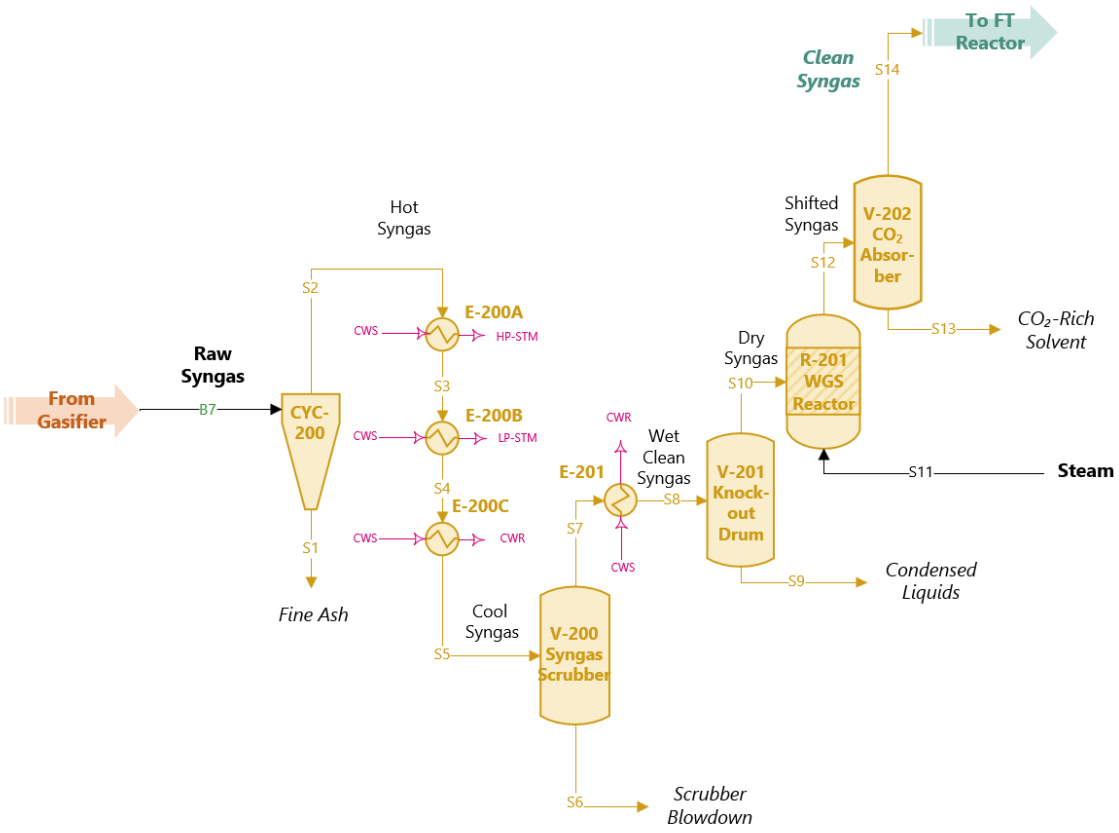


Figure 2: *Syngas Cleanup Process Flow Diagram*

Following the production of lean syngas, this is fed into the reactor section of the process ***Error! Reference source not found.***. This part of the process converts the conditioned syngas into long chain hydrocarbons producing waxes and middle distillates via the following chemical equations (Equation 3).

$$nCO + (2n + 1)H_2 \rightarrow C_nH_{2n+2} + nH_2O$$

$$CO + H_2O \rightarrow H_2 + CO_2$$

Equation 3*: Fischer-Tropsch Reactions*

The design basis will cover the optimal CO/H_2 ratio in the syngas feed. As previously mentioned, the FT-synthesis will occur in a fixed bed reactor at ~600°F and 290 psia with a 20wt.% iron supported catalyst. The FT reactor produces hydrocarbons with a molecular weight distribution which can be described by the Anderson-Schulz-Flory distribution.

$$\frac{W_n}{n} = 1 - \propto^2 \propto^{n-1}$$

Equation 4: *Anderson-Schulz-Flory distribution*

The catalyst was chosen due to Iron's high activity for long-chain paraffins in the desired SAF range of C_8-C_{16}. Alternatives not chosen include iron because it is less selective towards middle distillates, and nickel because of concerns of excessive methane formation. Key assumptions include those of the design basis.

High-pressure (HP) V-300 and low-pressure (LP) V-301 3-phase separators are added to separate heavy waxes (C20+) from liquid hydrocarbons and further separate liquid hydrocarbons into naphtha, middle distillates, and light hydrocarbons operating at ~300 psia and ~200 psia, respectively. Wax from the HP drum is sent to hydrocracking, and the fuel from the LP drum is vented. The LP drum *V-301* separates water and light hydrocarbons, removes inert gases, and vents flue gas. The light hydrocarbons are sent to the hydrocracking unit, and the upgrading processes follow.

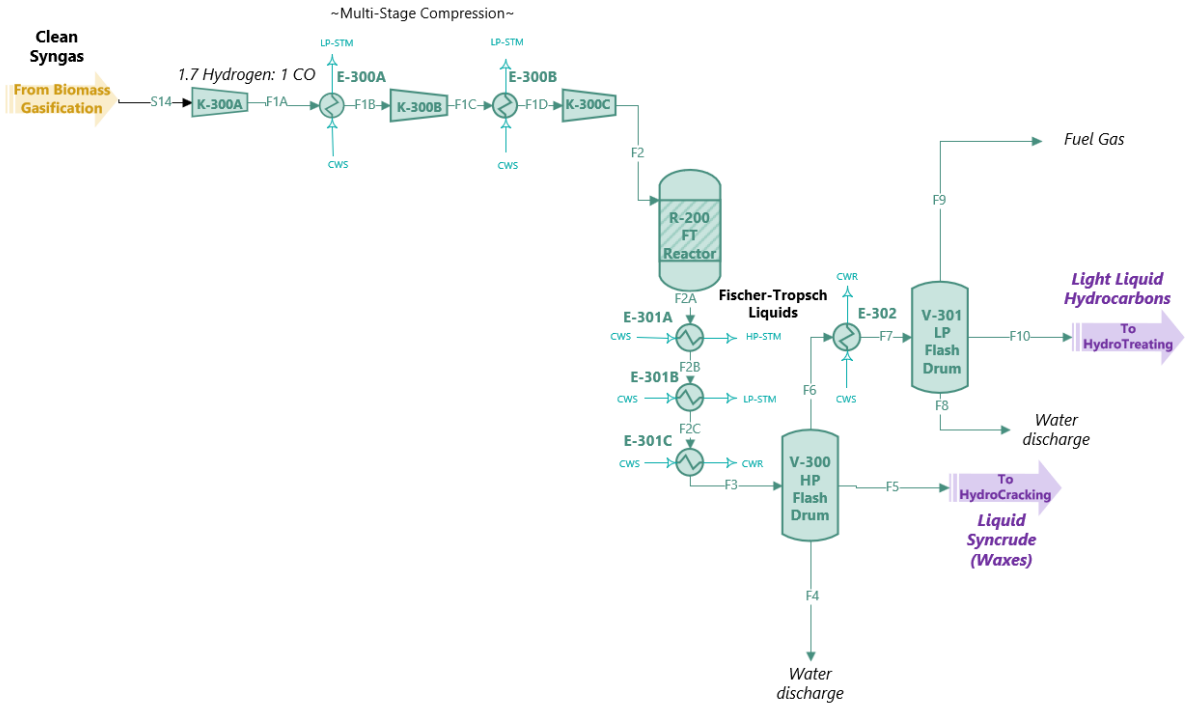


Figure 3: *Fisher-Tropsch Synthesis Process Flow Diagram*

The iron-catalyzed Fisher-Tropsch liquids which include linear paraffins, with a large fraction of heavy waxes (C_{20}^{+}) and a smaller fraction of middle distillates (C_9-C_{20}) must be sent to upgrading processes like hydrotreating and hydrocracking (**Error! Reference source not found.-Error! Reference source not found.**). While the hydrocarbons produced are sulfur and aromatics-free, they do not meet aviation fuel specifications due to excessive molecular weight and lack of isomerization. Hydrocracking and hydrotreating are required to crack FT waxes into jet fuel carbon range, isomerize linear paraffins into branched iso-paraffins to meet SAF freezing point limits, and produce co-product naphtha.

In the hydrotreating section *R-400* serves as the primary hydrotreating reactor for FT liquids ensuring hydrogenation of olefins and removal of trace oxygenates. The feed includes FT light liquid hydrocarbons, and a blue hydrogen recycle stream. Unreacted hydrogen is recycled

and supplied with purchased makeup hydrogen to comprise the hydrogen feed into the reactor. The feed is pumped and heated prior to entering *R-400*. Typical operating conditions include temperatures at ~450 °F and 900 psia. The flash drum *V-400* supplies unreacted hydrogen to the recycle stream and hydrotreated hydrocarbons into the first distillation column of the section. The primary fractionation column *C-400* separates light ends, naphtha, and unconverted waxes + jet fuel bottoms via atmospheric distillation. The column will typically operate at a top temperature of 150°C with a bottom temperature at 330-360°C at 1-2 bar. The final fractionation occurs in the finishing vacuum column *C-401* to separate SAF from heavier diesel-range hydrocarbons. Operating conditions here are like refinery kerosene fractionators and main products include SAF and diesel co-product.

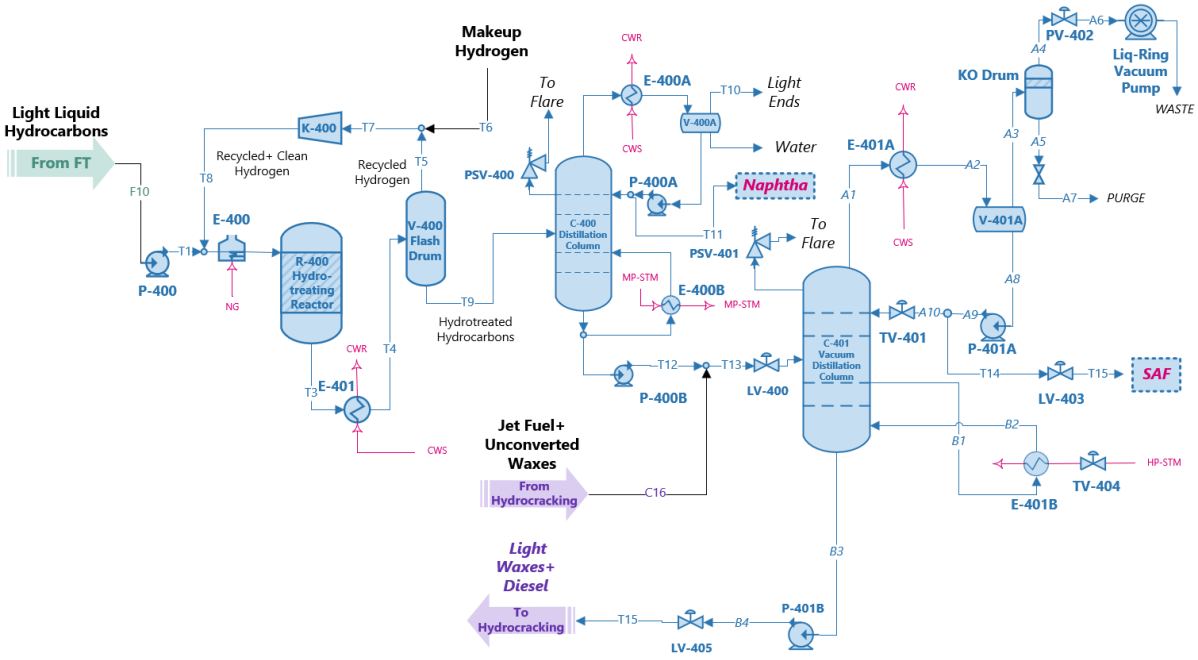


Figure 4: *Hydrotreating Process Flow Diagram*

The hydrocracking section (***Error! Reference source not found.***) serves to crack FT wax into lighter paraffins while simultaneously isomerizing products. Feed streams include light

waxes from hydrotreating (*T16*), liquid Syncrude from the FT synthesis, and makeup blue hydrogen (*C10*). A similar hydrogen gas recycle stream from the hydrotreating section is adopted here. The hydrocracker typically operates at 350°F and 900 psia. The conditions are chosen to promote isomerization and high wax conversion. The V-500 3-pase separating drum separates unreacted hydrogen, water discharge, and liquid cracked hydrocarbons at 120°F and 300 psia. The fractionation column *C-500* recovers Naphtha and light ends. Unconverted waxy material is recycled back to the hydrotreater to maximize SAF yield and overall carbon efficiency.

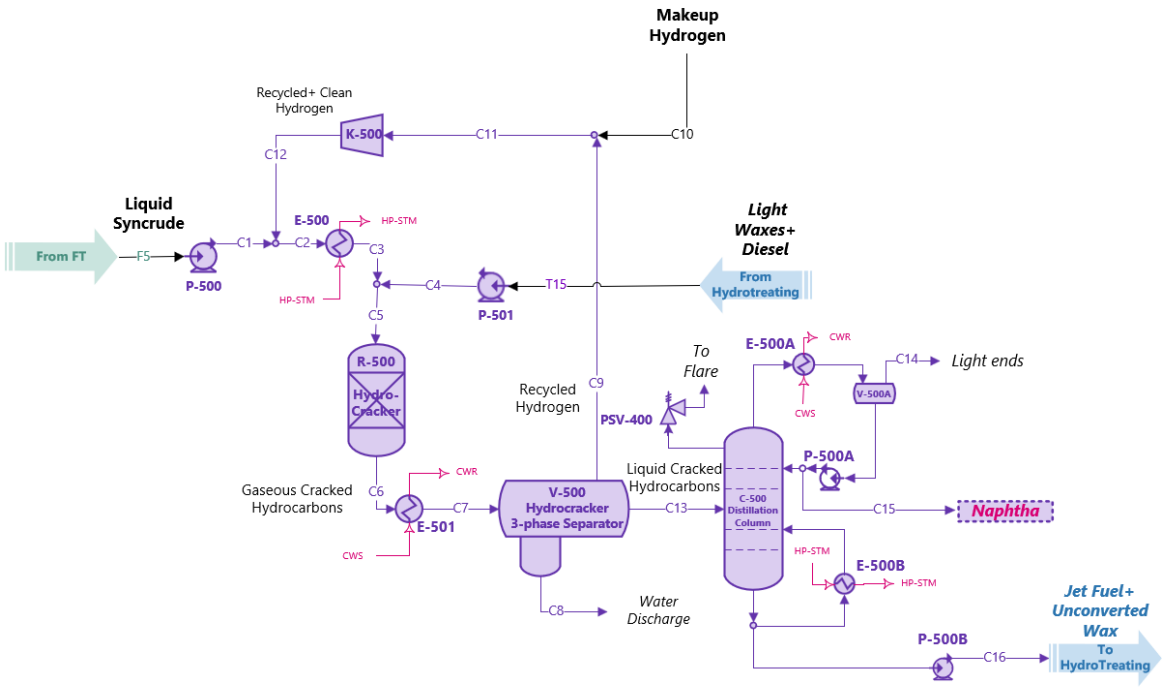


Figure 5: *Hydrocracking Process Flow Diagram*

Key assumptions in the upgrading section include that FT liquids contain negligible sulfur and nitrogen, eliminating the need for deep hydrotreating. The hydrogen is also assumed to be externally supplied from regional U.S Gulf Coat infrastructure. The hydrocracking severity is tuned to maximizing jet fuel range rather than gasoline.

2.2 Competing Processes

One of the main competing processes of Fischer-Tropsch process is the Hydro processed Esters and Fatty Acids (HEFA/HVO) process. Like the name suggests, hydro processed esters and fatty acids are the feed stock for this process with hydrotreated vegetable oil as another option. The first step of the process is pretreatment to remove any impurities. Pretreatment is crucial for SAFs especially since unlike traditional crude oil, it contains contaminants like phosphorus and metals. The pretreatment follows 5 steps: feedstock blending, straining, caustic treatment/water washing, impurity adsorption, and vacuum and filtration. Post pre-treatment hydrogenation and hydrodeoxygenation occur. Hydrogenation saturates any double bonds in the fatty acid chains, stabilizing them. Hydrodeoxygenation is where oxygen is removed via the creation of water (H_2O). Both processes are catalyzed, the first is typically by Pd/C, Pt/C, Ru, C or Raney NI for hydrogenation and Ni, Co, Mo, W and Al_2O_3 as typical catalysts for hydrodeoxygenation.

Following Hydrotreating, the product contains a large concentration of high carbon range materials C15- C18 paraffins, and this material is typically hydrocracked down to a useful range of carbons for SAFs (C8-C16). After the material is cracked there is a large concentration of straight chains that must be converted into branched isomers. The branched isomers have differing physical properties that are desirable for aviation fuel. These isomers control the fuel's ignition time, freezing point, viscosity, density, and flash point all of which must fall within aviation limits. After the isomers are created, they are separated via a high-pressure hot separator, cold high-pressure separator, and then fractionator columns. This leaves the products

jet-fuel, diesel or naphtha. After this, any remaining gas product has hydrogen recovered and is then recycled back into the hydrogenation reactor.

The HEFA process is not without its limitations and shortcomings. Primarily it is more economical for these plants to develop renewable diesel over SAFs. Secondly, feedstocks have existing uses such as food production. And finally, it is more rigid in its production of carbon ranges, typically producing only C8-C16. FT synthesis is able to be to be tailored to a desired range of carbons for various uses. The block flow diagram for a HEFA process is shown below.

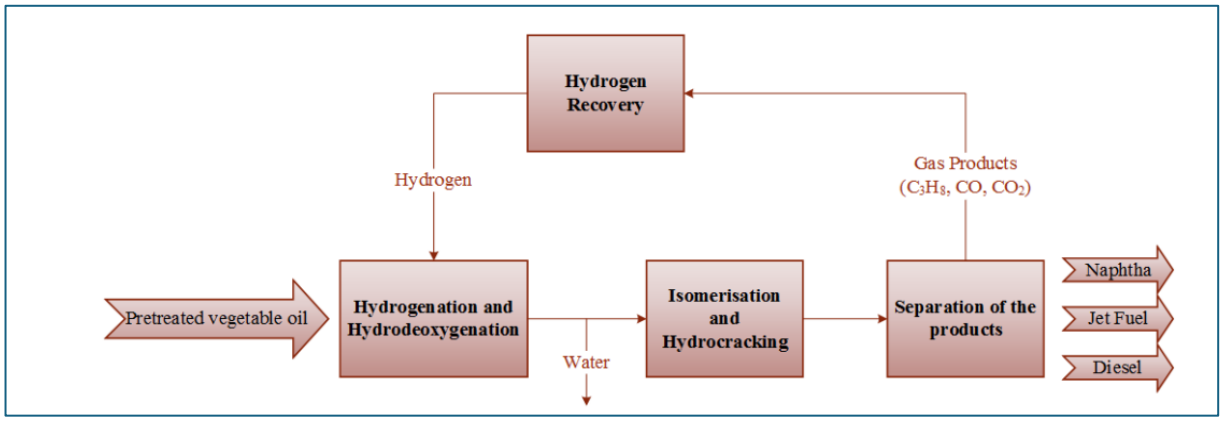


Figure 6: *HEFA Block Flow Diagram*

Another major competing process for Fischer-Tropsch synthesis is the alcohol to jet fuel (ATJ) route. The first step in the ethanol to jet process is the catalytic fueled dehydration of ethanol to ethylene. Typical catalysts for this process include HZSM-5 which is a zeolite and rare earth metals such as Lanthanum. HZSM-5 is the most common catalyst but is highly acidic, leaving it susceptible to coking. This results in a lower lifetime and large catalyst cost and regeneration rate. As a result, the ATJ process has a large catalyst cost and leaves room for improvements. Following the dehydration, the product is oligomerized. The olefins are then combined into larger carbon chains typically from the C8-16 range. Typical catalysts for this step

include zeolites and Ni-based. The typical choice for catalysts is Ni-ALSBA-15 due to a large conversion of ethylene at 95% and a high selectivity for the C5-C7 range.

Following the oligomerization, the material is often sometimes separated before it is hydrogenated, but this step is not necessarily done in this way. The last step of the process is hydrogenation, and this process is typically done through two separate reactors with two different catalysts. The first reactor typically targets the shorter hydrocarbons in the C8-C11 range while the second reactor achieves a higher conversion of the larger hydrocarbon chains. Typical catalysts for the hydrogenation step include Ni-C and Pt/AL2O3. Following hydrogenation, the product is separated via distillation.

Similarly, HEFAs/HVOS alcohol to jet fuel has its limitations. It also is limited in its carbon ranges of C8-C16. It does offer a more flexible feedstock than HEFA but lacks a large commercial scale. It is limited by alcohol availability in each area as alcohol is a high demand product for other industries. Overall FT synthesis offers a large amount of flexibility in product creation and a large amount of feedstock options. For these reasons the decision to go with an FT-based production pathway was made. The block flow diagram for ATJ process is shown below.

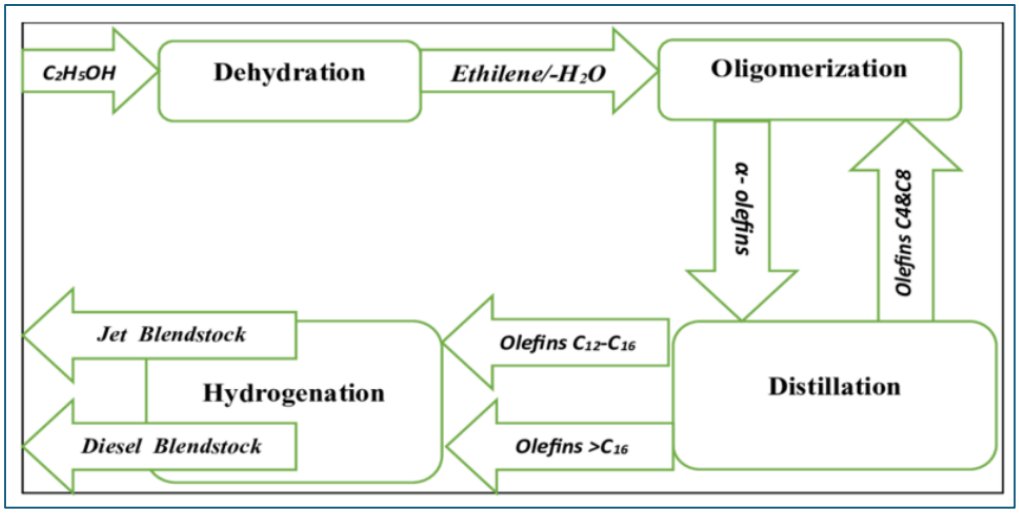


Figure 7: *ATJ Block Flow Diagram*

3. Process Control Analysis

3.1 Plant-Wide Control Strategy and Objective

To produce sustainable jet fuel safely and efficiently various controls must be put in place prior to plant operation. Each section highlighted in the BFD, Figure 18 has a different purpose therefore each section will be evaluated to apply the necessary controls. For this report only one unit, C-401, will be looked at in detail.

For the biomass gasification section of the process the main unit with operational concern is the gasifier. Variations with feed composition due to the inconsistent nature of biomass are the primary concern. Biomass will have varied particle sizes, moisture content and chemical composition that will impact syngas production. To combat disturbances in the feedstock, a typical process will control most of the following: feed flow, oxygen flow, ratio control, temperature and pressure. Since the output from the gasifier is temperature and pressure sensitive it is important to control these as well as possible. Deviations from the design

conditions can lead to an increase of tar formation, ash content and increased methane formation. All these deviations lead to a lower quality of syngas and therefore lower quality SAF.

In the syngas cleanup section of the process the main topics of concern are syngas composition and tar removal. The most important unit operation in this section of the process is the cyclone. The cyclone will remove the ash from the system ensuring the rest of the undesired liquids can be removed in the subsequent flash drums producing the desired H2:CO ratio. Typical controls for the cyclone include inlet flow, pressure, temperature. These control variables directly correlate to the degree of separation in the cyclone and are therefore extremely important to maintain consistent operation. Another major area of concern for this area of the process is the water gas shift reactor. Since the water gas shift reaction is exothermic temperature control is important for safe operation. This alongside controlling the flowrate of steam into the reactor will for optimal syngas production.

In the FT section of the process the main area for concern is the Fischer Tropsch reactor itself. By far the most important control variable within the FT reactor is the temperature. Since the FT reaction is exothermic the temperature must be closely regulated to prevent a runaway reactor and ensure consistent products. The reactor pressure also has a strong correlation to the product distribution and conversion which must fall in a specific range for the system to operate efficiently. The system is designed to function primarily with a large number of heavier hydrocarbons and varying the temperature will change the carbon distribution greatly affecting operation.

In the hydrocracking section of the process the main areas of concern are the hydrocracker and the distillation column downstream of it. Since the hydrocracker is an overall exothermic process, temperature control is important for both operational safety and reaction

kinetics. Pressure control alongside make-up hydrogen flow rate are other major necessary controls. The distillation column will have controls similar to the fractionation column to be discussed in the control loop section.

In the hydrotreater section of the process the areas of concern are the hydrotreater reactor and its associated temperature, pressure, flow and conversion alongside the operation of both distillation columns. The first distillation column C-400 will require temperature, pressure and level controls to ensure constant production of on-spec naphtha and continuous separation of SAF.

3.2 Control loops for Vacuum Distillation Column *(C-401)*

The fractionation vacuum column C-401 is chosen to design full control loop because it is where the main product from the system is obtained and therefore control of the unit is of the utmost importance. The aim of the control of this unit is to maintain the product, synthetic paraffinic kerosene, on spec while also maintaining safe operating conditions. This is

accomplished through 5 control loops outlined below.

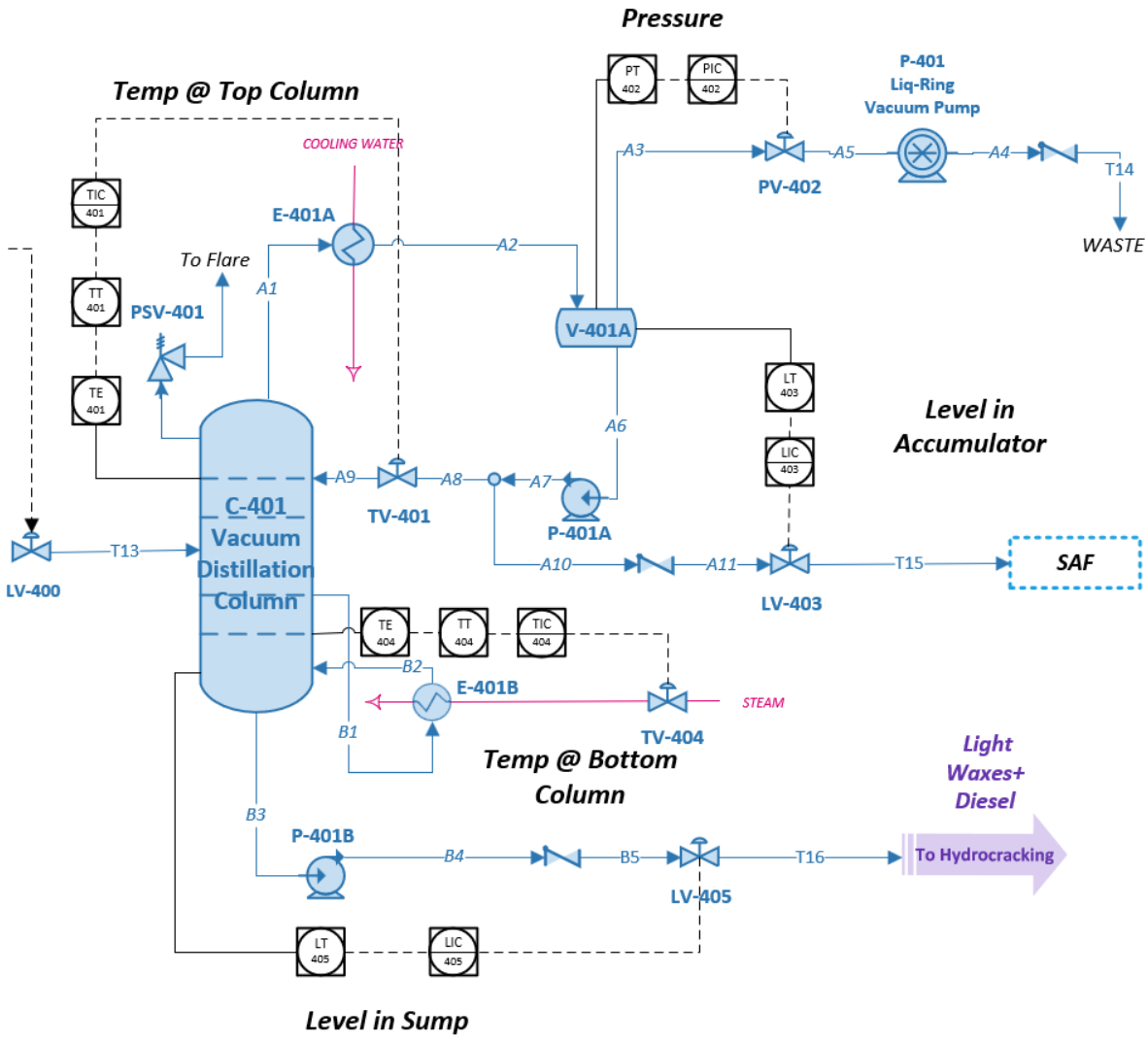


Figure 8: *Control Loop for C-401 before HAZOP Analysis*

The first control loop is designed to control the temperature near the top of the column. The temperature is recorded a few trays into the tower and if it deviates from the set-point the controller will send a signal to TV-401 to adjust the flowrate of the reflux entering the tower. If the temperature is lower than the set-point, the controller will reduce the reflux flowrate and if it is higher than the set-point, then it will increase the reflux flowrate.

The second control loop is designed to control the pressure in the column. The pressure is measured in the reflux-accumulator V-401A. If the pressure is higher than the set point the controller sends a signal to PV-402 to increase the flowrate of the gaseous stream allowing for a higher vacuum to be pulled. If the pressure is too low the controller will reduce the flowrate of the gaseous stream to reduce the vacuum. The low-pressure scenario is not as major of concern as the column is designed for full vacuum.

Next the third control loop is designed to maintain the level in the reflux-accumulator by manipulating the distillate flowrate. If the level is too high the controller sends a signal to LV-403 to increase the distillate flow rate. If the level is too low the controller sends a signal to reduce the distillate flowrate to return the level to its setpoint.

Additionally, the fourth control loop is designed to control the temperature in the reboiler. The temperature is once again measured a few trays from the bottom to help prevent rapid fluctuations. This temperature is measured and if it is higher than the set-point the controller sends a signal to TV-404 to reduce the steam flow rate. If the temperature is lower than the set-point, the controller sends a signal to increase the steam flow rate.

Finally, the fifth control loop is designed to maintain the level of the sump tank by manipulating the bottoms flow rate. If the sump level is too high the controller will send a signal to LV-405 to increase the bottoms flowrate. If the level is too low the controller will send a signal to decrease the bottoms flowrate to reach the set-point. These control loops are preliminary and will be modified post HAZOP analysis with additional safeguards.

4. Process Safety and Environmental Analysis

4.1 Chemical and Operational Safety

4.1.1 Material Safety

Material safety must be considered for each component present in the entire process. A general material safety description will be provided for each major section of the process as shown in the BFD in *Figure 18***:** Final Block Flow Diagram along with a table highlighting individual components.

Biomass Gasification:

The biomass gasification section deals with the handling of woody biomass which can be susceptible to spontaneous combustion and must be handled and stored properly. The woodchips should be maintained in a low-dust cool environment. Respiratory protection against dust must be implemented when handling the biomass. Once the biomass is gasified, the resulting syngas must also be safely handled and its component's MSDS sheets should be reviewed. The components of syngas, carbon monoxide and hydrogen, are also reviewed.

Syngas Cleanup:

Once the syngas is ready to be sent to the cleanup section, a carbon dioxide absorber will ensure that a low concentration of CO_2 remains in the final syngas stream. The waste stream of the absorber will be a CO_2 rich solvent that must be disposed of adequately.

FT Synthesis:

Clean syngas will undergo various catalytic reactions to yield Fisher-Tropsch liquid hydrocarbons which are highly flammable and should maintain as far as possible from any possible ignition sources. This means installing ground containers to prevent static, explosion proof equipment, and following OSHA 1910.106 standards for flammable liquid storage and

handling. The condensed liquids from the various cleanup units should be disposed of properly to prevent any contaminants coming into contact with the public.

Hydrotreating:

Following the low-pressure drum in the synthesis section of the process, light liquid hydrocarbons enter the hydrotreating section of the process. Similarly to the FT section, 1910.106 regulations are followed along with additional regulations for hydrogen like the 1910.103 standards.

Hydrocracking:

Comparably to the hydrotreating section, in the hydrocracking section the same regulations will be followed. OSHA also explicitly states that a process containing 10,000 lbs. of Hydrogen is PSM-covered as a flammable gas process. Hydrogen streams should be handled away from possible ignition sources.

4.1.2 Components

The individual components present in the entire process can be summarized in the table below. The following data was obtained from each component's MSDS sheet. The flammability limits are the highest and lowest concentrations in air that can propagate a flame if ignited at normal pressure and temperature. For substances that are not flammable, this property is not applicable like in the case of water. The boiling point is the temperature at which the substance can enter the liquid phase. While most substances have an agreed value, more complex hydrocarbons, like the products of this process, boil over a range of temperatures. The flash point is the lowest temperature of a liquid that can give off substantial flammable vapors. This chemical property can only be applied to substances that are liquid at standard conditions. The

autoignition temperature is the lowest temperature at which a substance can spontaneously ignite without an external flame or spark. The toxicity of a substance is qualified as being potentially harmful to humans. The heat of combustion is the amount of heat produced when a substance undergoes complete combustion with oxygen. If the substance is not combustible, then the heat of reaction is given instead. All these parameters are important and must be accounted for when designing and planning for materials safety.

Hydrogen is an extremely flammable gas, classified as a level 4 fire hazard. The main health hazard is that hydrogen can quickly displace oxygen in air as it is a simple asphyxiant. It is non-toxic but dangerous due to its flammability and ability to lower oxygen levels untraceably.

Carbon monoxide is colorless and odorless highly toxic gases. It poses a level 3 health risk as it is toxic if inhaled and may cause damage to organs through prolonged exposure. It is also highly flammable, being a level 4 fire hazard, and should be kept in a cool dry area away from ignition sources. The OSHA Permissible Exposure Limit, PEL, is 50 ppm for workers.

Water is nonhazardous but when heated it can cause potential burns if it comes into contact with skin. Water and steam pose no health hazards but should be handled with care.

Carbon dioxide is an odorless, colorless asphyxiant.

Oxygen is a colorless, odorless gas at most temperatures and presents a combustion hazard especially at a large purity. Oxygen acts as an oxidizer and at the high concentrations necessary by the process it is highly combustible. It is important to keep flames and other ignition sources away from the feed oxygen streams.

Nitrogen is an inert gas that does not burn or support combustion. It poses a level 3 health risk because it is not poisonous but can be a simple asphyxiant. Liquid Nitrogen is also extremely

cold and can cause instant frostbite upon contact with skin. Proper ventilation and cryogenic PPE when handling nitrogen is essential.

Hydrogen Sulfide is only present in trace amounts throughout the plant, but safety measures should be taken regardless. It is a colorless, flammable gas that poses a

Methane is also highly flammable and poses a suffocation risk. It must be handled in well-ventilated areas away from ignition sources.

Ethane is highly flammable gas that poses explosion threats and must be kept in cool ventilated areas away from ignition sources

Propane is a colorless gas that is primarily a fire and asphyxiation hazard. It poses a health rating of 2 and a fire rating of 4, making it not as toxic but can ignite very easily. It should be kept in well-ventilated areas away from heat, sparks, flames, and other ignition sources. OSHA lists propane with a PEL of 1000 ppm as an 8-hour TWA and also notes it is treated as a simple asphyxiant, so oxygen deficiency is a major concern in confined or poorly ventilated spaces

Jet Fuel is a combustible petroleum liquid with a petroleum odor that is best treated as both a flammable liquid hazard and an inhalation / skin contact hazard. OSHA's kerosene (jet fuels) entry gives it an NFPA health rating of 2 and an NFPA fire rating of 2

Diesel fuel is a combustible petroleum liquid that presents more of a combustible liquid and chronic exposure hazard than an extreme flash-fire hazard under normal conditions. OSHA's diesel fuel page lists an NFPA health rating of 1 and an NFPA fire rating of 2.

Naphtha is a volatile petroleum liquid mixture that is both a flammability hazard and an inhalation hazard. OSHA's VM&P naphtha entry gives it an NFPA health rating of 1 and an NFPA fire rating of 3. It can ignite much easier than diesel or jet fuel

Table 1: *Component Chemical Properties*					
Component	Flammability Range	Boiling Point (°C)	Flash Point	Autoignition Temperature (°C)	Heat of Combustion or Reaction kJ/mol
H_2	4.1-74.2	-252.88	n/a	500	-286
CO	12.5-74.2	-191.5	n/a	609	-283
H_2O	n/a	100	n/a	n/a	n/a
CO_2	n/a	-78.5	n/a	n/a	-393.5
O_2	n/a	-182.96	n/a	n/a	-418
N_2	n/a	-195.8	n/a	n/a	941
H_2S	4.3-45	-60.3	n/a	260	519
CH_4	5.3-14	-161.5	n/a	540	802
C_2H_6	3-12.5	-88.6	n/a	515	1419
C_3H_8	2.3-9.5	-42.25	n/a	450	2029
Jet Fuel	0.7-5.6	150-250	37.8	210	~7000
Diesel	6.0-13.5	170-360	>38	>200	~8090
Naphtha	1.1-7.5	30-200	23	225	~3680

4.1.3 Operational Safety

For the general operation safety of, the plant 29 CFR 1910 standards will be followed

Table 2: *Operational Safety Guidelines*		
Unit	Safety Concerns	Prevention and Mitigation
Biomass Dryer	Fire or Explosion Conditions	Temperature monitoring and high temperature shutdown.
	Dust from woodchips	Install dust suppression, proper ventilation, and enforce proper respiration protection
Air Compressor	Overpressure	Discharge pressure protection
	Seal leaks	Routine seal inspections

Biomass Gasifier	High Temperatures	Temperature monitoring and high temperature shutdown.
	Flammable gasses	Gas leak detection
Water gas shift reactor	Temperature rise	Tight temperature monitoring across bed
	Overpressure	Pressure relief valves
	CO Leaks	Use CO detection around reactor area
CO_2 Absorber	CO_2-rich solvent leaks	Provide secondary containment for possible solvent leaks
	Column Flooding	Differential Pressure monitoring
Syngas Compressors	High-pressure syngas leaks	Pressure monitoring
	Flammable Gases	CO detectors
	Seal failure	Seal monitoring and leak detection
Fischer-Tropsch Reactor	Highly Exothermic Reaction	Temperature measurements across reactor
	CO and H_2 Leaks	Feed control, pressure relief to safe disposal system
	Runaway temperatures	High-high temperature emergency shutdown
High Pressure Flash Drum	High Pressures	Pressure relief and shutdown system, level control
Low Pressure Flash Drum	Flammable vapor released	Route off gas safely
Hydrotreating Reactor	High pressure hydrogen and hydrocarbon mixture	High-high temperature and pressure trips
	Exothermic reaction and hotspot formation	Temperature monitoring across bed
	Flammable gas leak and fire risk	Relief to safe disposal system
Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Flooding	DP monitoring
Vacuum Distillation Column	Vacuum Loss	Vacuum monitoring and alarms

Hydrocracker	High pressure hydrogen	High-high pressure alarm and trips
	Strong exothermic reaction-hotspot formation	Multiple temperature measurements
	Hydrogen leak and fire risk	Relief to safe disposal
3-phase separator	Hydrogen rich gas release	Closed vent handling
Recycle Hydrogen Compressor	Hydrogen leaks	Seal monitoring and leak detection
	Overheating	High temperature shutdowns
	Fire and explosion risk	Flammable gas detection
Hydrotreater Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Over head vapor release	Safe overhead vent handling

4.2 Process Safety and Environmental Analysis

4.2.1 Air Emissions

Air emissions from the process primarily originate from gas-phase separation and upgrading units, with carbon dioxide (CO_2) as the dominant component formed during gasification and downstream reactions. Additional emissions include flue gas from flash drum operations and light hydrocarbon ends from distillation columns in the hydro processing section. CO_2 is not classified as a hazardous air pollutant but is a significant greenhouse gas, while flue gas streams are generally non-hazardous but may contain trace combustion byproducts. The light ends streams consist of low-molecular-weight hydrocarbons that are flammable, posing handling and safety risks, though they are not considered highly toxic. These emissions and their sources are summarized in Table 3 below.

***Table 3*: Air Emissions and Sources**

Material	Amount (tonne/op-year)	Source
Carbon Dioxide	149,300	Gassifier

Flue Gas	309,482	Low Pressure Flash Drum
Light Ends	12,135	Hydrotreater distillation column
Light Ends	4,950	Hydrocracker Distillation Column

4.2.2 Liquid Waste Treatment

Liquid waste streams arise mainly from biomass processing, gas cleanup, and separation operations, including water from biomass moisture, condensed process water, tar from the gasifier, and scrubber blowdown from syngas cleaning. While the majority of these streams are water-based and not inherently hazardous, they may contain dissolved organics, particulates, or contaminants that require treatment prior to discharge. The tar stream is considered hazardous due to its content of heavy hydrocarbons and potentially toxic compounds, and scrubber blowdown may also require treatment depending on contaminant levels, making proper liquid waste management essential for environmental compliance. These liquid waste streams and their sources are summarized below.

Table 4: *Liquid Waste and Sources*

Material	Amount (tonne/op-year)	Source
Water	712,500	Biomass Moisture
Tar	9,151	Biomass Gasifier
Scrubber Blowdown	19,800	Syngas Scrubber
Water Discharge	35,550	HP Flash Drum
Water Discharge	6,280	3-phase Separator

4.2.3 Solid Waste Treatment

Solid waste generated in the process is primarily composed of ash from biomass gasification, along with minor particulates removed during gas cleanup. This ash consists mainly of inorganic minerals originally present in the biomass feedstock and is generally non-hazardous, provided it does not contain elevated levels of heavy metals or toxic contaminants. Although typically suitable for landfill disposal or potential reuse, the ash may require characterization to confirm compliance with environmental regulations. These solid waste streams and their sources are summarized below.

Table 5: *Solid Waste and Sources*

Material	Amount (tonne/op-year)	Source
Ash	20,176	Syngas Cleanup Cyclone

4.3 Hazard and Operability Study *HAZOP* for Vacuum Column *C-401*

A comprehensive hazard and operability study was conducted for the vacuum column C-401 as shown in the table below. The software OPEN PHA was used to conduct the study. Deviations were evaluated for the severity of their consequences and appropriate safeguards and recommended actions were applied.

Table 6: *Hazard and Operability Study*

Main Vessel- ***Vacuum Distillation Column C-401***

Intention- to separate useful jet fuel hydrocarbons from light waxes and diesel hydrocarbons via vacuum distillation

Deviation	Consequence	Cause	Safeguard	Recommended Action
MORE Pressure	Vessel mechanically fails due to overpressure	TV-404 fails open, excess steam flow provides more heat to the reboiler than	- PT-402 readout in control room.	- Add PAH-402 - Add TAH-401

	and potential fatality.	the overhead condenser system can remove. SAF components no longer condense, vacuum pump not rated for higher flow of vapor, pressure builds to MAWP, vessel failure, loss of process containment and potential fatality.	- PSV-001 safety relief valve. - TT-401 readout in control room	
LESS Pressure	Vessel fails due to high vacuum, collapse	PV-402 fails open, excess flow through K-401 vacuum pump, pressure drops below vacuum set-point, potential to fail column, potential fatality.	- Column is designed for full vacuum (FV).	- Add PAL-402
LESS Flow	No new consequences see high pressure	TV-401 fails closed, loss of reflux, excess heat to column, potential for overpressure to MAWP, vessel failure, loss of process containment and potential fatality.	- PT-402 readout in control room. - PSV-001 safety relief valve. - TT-401 readout in control room	- Add PAH-402 - Add TAH-401
MORE Flow	No new consequences, see high pressure	TV-404 fails open, causing excessive vapor traffic through the column	- TT-401 readout in the control	- Add PAH-402 - Add-TAH-401

			- -PSV-001 safety relief valve	
LESS Level	See high pressure for ultimate consequence. Evaluate seal fire here.	LV-403 fails open, excess material pumped from V-401A, loss of level, pump P-401A runs dry, equipment damage, potential seal fire, potential injury. Loss of reflux also occurs, leading to high pressure.	- TT-401 readout in control room	- Add TAH-401 - Install redundant level devices with alarms. - Install double mechanical seal with seal routed to flare, high pressure alarms on seal pot.
MORE Level	No new consequences, see low pressure	LV-403 fails closed, material accumulates in V-401A and be enrtrained towards K-401, potential damage to P-401A, loss of vacuum pressure	- PT-402 readout in the control room - KO drum	- Add LAH-403 - Consider adding LAHH to isolate K-401
LESS Temperature	No new consequences, see low pressure	TV-404 fails closed, TIC-404 reads low, or TT-404 reads high. TV-401 fails open	- PT-402 readout in the control room - TT-401	- Add TAL-401
MORE Temperature	No new consequences, see high pressure	TV-404 fails open, TIC-404 can read high, or TT-404 can read low,	- PT-402 readout in the control room	- Add TAH-404 - Consider TAHH to

		allowing excess steam to E-401B TV-401 fails closed, loss of reflux, excess heat to column, potential for overpressure to MAWP, vessel failure, loss of containment and potential fatality	- TT-401 readout in control room - PSV-400 pressure relief valve	isolate steam - Consider adding LAL to cooling water flow on E-401A

After conducting the HAZOP study, additional safety measures were added to the final P&ID shown below in ***Error! Reference source not found.***. Most notably, a safety pressure relief valve was added to the column along with alarms to indicate high and low deviations from various setpoints. These controls were added to the column to ensure safe operation.

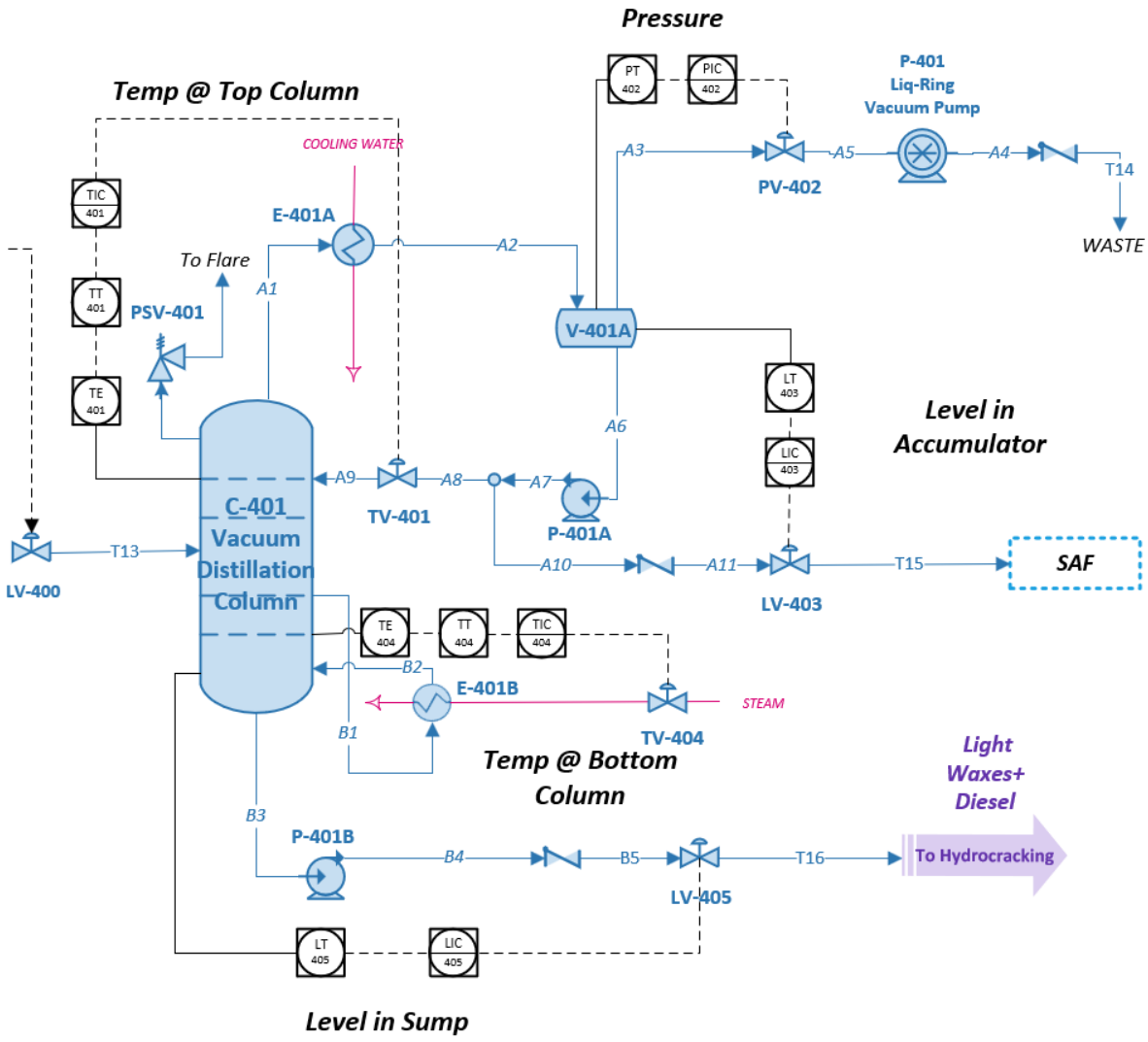


***Figure** 9: Final P&ID for Vacuum Fractionation Column C-401*

5. Process Economic Analysis

An economic analysis was conducted to evaluate the financial feasibility of the proposed biomass-to-SAF process. This analysis includes estimates of both capital investment and operating costs associated with the facility. Capital costs were determined using a combination of Aspen Process Economic Analyzer results and manual equipment cost estimation methods, while operating costs were estimated based on raw material consumption, utilities, catalyst

replacement, and fixed operating expenses. Using these cost estimates, a discounted cash flow analysis was performed to determine the discounted cash flow rate of return (DCFROR) for the project. The DCFROR was evaluated using both 5-year and 7-year MACRS depreciation schedules, and a sensitivity analysis was conducted to examine how variations in key economic parameters affect the overall profitability and feasibility of the process.

5.1 Capital Costs

The inside battery limits (ISBL) capital cost for the process was initially estimated using equipment cost results generated by Aspen. However, Aspen does not always provide accurate cost estimates for all equipment, particularly for custom equipment, reactors, or units outside its costing database. Therefore, major process units and any equipment that Aspen did not cost were manually estimated. These manual estimates were performed using the cost correlations provided in Table 7.2 of Towler and Sinnott, as well as by comparing the equipment size and duty to similar units reported in literature and scaling costs when Aspen estimates appeared unrealistic. These steps produced unrealistically low capital costs, so FT plants of similar scale were also taken into consideration when estimating the capital costs. The ISBL cost breakdown by section is shown in the figure below.

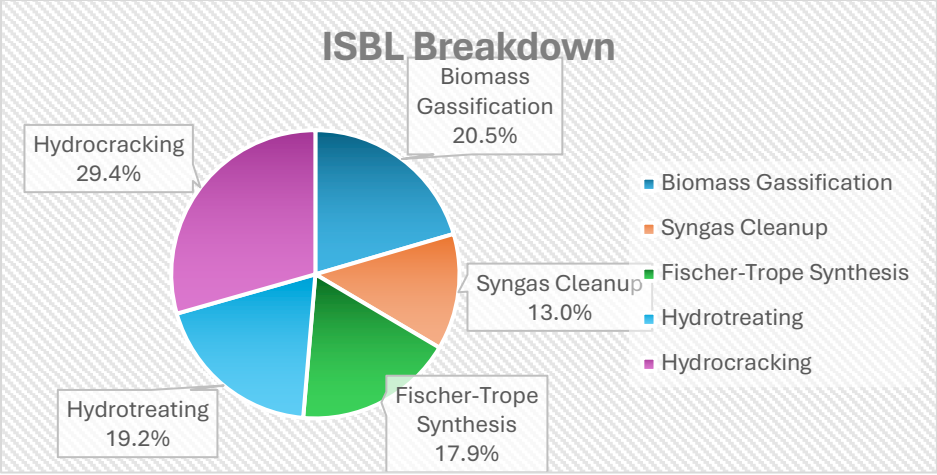


Figure 10: *ISBL Breakdown by BFD Section*

After estimating the base equipment costs, a material of construction correction factor F_m was applied to account for equipment constructed from alloys other than carbon steel. The material of construction for each unit is listed in Appendix V, and the corresponding F_m values were taken from Towler and Sinnott.

Next, a Chemical Engineering Plant Cost Index (CEPCI) correction was applied to adjust Aspen equipment costs to more recent cost conditions. Aspen's costing database corresponds to the year 2022 (CEPCI = 816). Costs were updated to 2025 conditions (CEPCI = 805) using the CEPCI ratio. However, the majority of major equipment costs were obtained from manual calculations based on more recent references, so this correction had only a minor impact on the overall capital estimate.

A location factor is typically applied to account for differences in construction costs between geographic regions. Since the plant is assumed to be located in Louisiana along the U.S. Gulf Coast, the standard Gulf Coast reference location was used and the location factor was taken as 1.0.

Once the ISBL cost was determined, additional capital costs were estimated using standard percentage factors. The following fractions of ISBL were applied:

- Outside Battery Limits (OSBL) = 0.40 × ISBL
- Engineering and Construction Costs (ECC) = 0.10 × ISBL
- Contingency (X) = 0.10 × ISBL
- Working Capital (WC) = 0.15 × ISBL

These factors account for offsite infrastructure, engineering and construction expenses, project uncertainty, and initial operating cash requirements. The resulting capital cost breakdown, including ISBL, OSBL, ECC, contingency, and working capital, is summarized in the table below. The sum of these components gives the total capital investment required for the plant.

Table* 7*: Capital Cost Summary	
Cost Type	Cost ($MM)
Inside Battery Limits (ISBL)	560.3
Outside Battery Limits (OSBL)	224.1
Engineering and Construction Costs (ECC)	56.0
Contingency (X)	1560.0
Working Capital (WC)	84.1
Total Capital Cost (TCC)	980.6

5.2 Operating Costs

Operating costs were estimated by separating the variable cost of production (VCOP) and the fixed cost of production (FCOP). An 8000-hour operating year for the plant was assumed in

all calculations. Variable costs include raw materials, utilities, catalyst replacement, and wastewater treatment. Fixed costs include labor, maintenance, and plant overhead. The cash cost of production (CCOP) is then determined as the sum of VCOP and FCOP:

$$CCOP = VCOP + FCOP$$

Equation 5: *Cash Cost of Production*

The calculated operating costs for the proposed process are summarized in the following subsections.

5.2.1 Variable Cost of Production (VCOP)

The variable cost of production consists of the costs of raw materials, utilities, catalysts, and wastewater treatment. These costs scale directly with plant throughput and therefore represent the largest portion of the operating expenses.

Utilities required for the process include cooling water, natural gas, and electricity. High-pressure and low-pressure steam are also used in the process. However, the plant generates significant quantities of steam through heat recovery in several heat exchangers.

The water–gas shift (WGS) reactor requires steam for operation. Rather than purchasing steam, the process uses steam generated internally through heat integration. Since the process produces more steam than is required, the excess steam offsets the cost of steam generation. Because the facility does not include infrastructure to export steam, the steam is assumed to be used internally at no net cost.

Utility prices used in the analysis where provided by professor Michael Caracotsios, which are:

- Cooling water: $0.354/GJ

- Natural gas: $6/GJ
- Electricity: $0.0561/kWh

The total utility consumption and associated annual costs are summarized in the table below.

***Table** 8: Utility Consumption and Cost Summary*			
Utility	**Consumption**	**Unit**	**Annual Cost ($/yr)**
Cooling Water	150.5	GJ/hr	4.3×10^5
Natural Gas	130	GJ/hr	6.2×10^6
Electricity	1.83×10^4	kWh	8.2×10^6
Steam (HP & LP)	Generated internally	—	—
Total Utility Cost	**—**	**—**	**$14.8 MM/yr**

Raw material costs represent the largest component of operating expenses. The primary feedstocks required for the process are biomass, oxygen, and green hydrogen.

Biomass provides the carbon source for gasification, oxygen is required for partial oxidation in the gasifier, and hydrogen is consumed in downstream upgrading reactions including hydrotreating and hydrocracking. The unit prices are sourced from *EIA* and *Business Analytiq*. The raw material consumption and annual costs are reported on the table below.

***Table** 9: Raw Material Consumption and Cost*			
Raw Material	**Flow Rate**	**Unit Price**	**Annual Cost ($/yr)**
Biomass	5.2×10^5 lb/hr	$0.03/lb	1.26×10^8

Hydrogen	1.6×10^3 kg/hr	\$7.5/kg	9.55×10^7
Oxygen	5.0×10^3 kg/hr	\$0.11/kg	4.4×10^6
Total Raw Material Cost	**—**	**—**	**\$225.7 MM/yr**

Catalysts are required in the Fischer–Tropsch reactor, hydrotreating reactor, and hydrocracking reactor. The Fischer–Tropsch reactor uses a low-temperature iron catalyst promoted with Fe–Cu–K. Iron catalysts are commonly used for Fischer–Tropsch synthesis with biomass-derived syngas because they tolerate lower H_2/CO ratios and exhibit water–gas shift activity, which helps adjust the hydrogen balance during synthesis.

The hydrotreating and hydrocracking catalyst systems were selected based on the configuration described in the Rentech upgrading technical report, which outlines the catalyst loading schedule used for upgrading Fischer–Tropsch liquids. However, the detailed catalyst compositions are proprietary, so representative catalyst prices were used for economic estimation. The hydrotreater contains multiple catalyst layers consisting of a guard bed followed by hydrotreating catalysts. The guard bed, Crystaphase CatTrap 30, removes contaminants that could deactivate downstream catalysts. The remaining catalyst layers (HC-DM, HC-T, and AS-200LT) perform hydrogenation and stabilization reactions prior to hydrocracking.

The hydrocracker also contains multiple catalyst beds with different functions. The first section contains hydrogenation catalysts (HC-DM and HC-T), which saturate intermediate species and stabilize the feed. The second section contains HC-205LT, which provides the stronger cracking functionality required to convert heavier Fischer–Tropsch products into lighter

liquid fuels. This split between hydrogenation and cracking catalyst beds is typical of industrial hydrocracking reactors and is consistent with the configuration described in the Rentech report.

The catalyst inventory for each reactor was estimated using the reactor volume and a typical catalyst bulk density. As an example, the catalyst requirement for the Fischer–Tropsch reactor was estimated using the relationship

$$m_{cat} = V_{reactor}\rho_{bulk}$$

Equation 6: *Catalyst Requirements*

Assuming a reactor catalyst volume of 18.4 m³ and an iron catalyst bulk density of 800 kg/m³, the catalyst inventory becomes

$$m_{cat} = (18.4 m^4)(8000\ kg/m^3) \approx 14{,}750\ kg$$

Using an assumed catalyst price of $12/kg, the catalyst cost is 177,000$. Assuming a two-year catalyst lifetime, the annual catalyst replacement cost becomes

$$\frac{177{,}000}{2} = 88{,}500\ \$/yr$$

A similar procedure was used to estimate the catalyst inventory for the hydrotreating and hydrocracking reactors using their respective reactor volumes and catalyst packing densities. Because the hydrocracker contains separate hydrogenation and cracking catalyst sections, the total hydrocracker catalyst inventory was divided between these two catalyst types. FT Catalyst prices where acquired from *Trading Economics*, while the rest where estimated based on material since specific information about hydrotreating and hydrocracking catalyst could not be found.

The resulting catalyst inventories and annual replacement costs are summarized in the table below.

***Table** 10: Catalyst Inventory and Annual Replacement Cost*					
Reactor	Catalyst System	Catalyst Mass (kg)	Cost ($/kg)	Lifetime (yr)	Annual Cost ($/yr)
FT Reactor	Fe–Cu–K promoted iron catalyst	14,750	12	2	88,500
Hydrotreater	Guard + hydrotreating catalyst system (CatTrap 30, HC-DM, HC-T, AS-200LT)	7,060	25	4	44,125
Hydrocracker	Hydrogenation catalyst section (HC-DM / HC-T)	2672	45	4	30,000
Hydrocracker	Cracking catalyst section (HC-205LT)	6232	90	4	140,000
Total Annual Catalyst Cost	—	—	—	—	**300,000**

Wastewater generated from process condensate streams and cooling operations was estimated to be 2,500 gallons per hour. Wastewater treatment costs were estimated using an industrial treatment cost of $2 per 1000 gallons, resulting in an annual wastewater treatment cost of approximately $40,000 per year.

Solid waste disposal costs were estimated from the ash and tar byproduct streams. Ash was treated as a nonhazardous solid waste and costed using an average U.S. landfill tipping fee of approximately $62/ton. Tar was assigned a higher disposal cost of $300/ton as a preliminary estimate to reflect the more complex handling typically associated with heavy organic residues. Based on production rates of 500 lb/hr ash and 2,500 lb/hr tar, and assuming 8,000 operating hours per year, the total annual solid waste disposal cost was estimated to be approximately $3.13 million per year.

The total variable cost of production was calculated by summing up the individual variable cost components:

$$VCOP = C_{raw} + C_{utilities} + C_{catalyst} + C_{wastewater} + C_{solids}$$

Equation 7: *Variable Costs of Production*

$$VCOP = 225.8 + 14.8 + 0.175 + 0.040 + 3.13$$

$$VCOP = 243.8 \text{ \$MM/yr}$$

5.2.2 Fixed Cost of Production (FCOP)

The fixed cost of production (FCOP) represents operating expenses that do not vary significantly with plant throughput. These costs include operating labor, supervision, maintenance, plant overhead, and taxes and insurance. The FCOP was estimated using the standard cost factor approach described in Towler and Sinnott, where several cost components are estimated as fractions of labor cost or fixed capital investment.

The plant was assumed to require five operators per shift operating over three shifts, with an annual salary of $70,000 per operator. Additional costs for supervision, direct overhead, maintenance, plant overhead, and taxes and insurance were then estimated using typical percentage factors reported in Towler and Sinnott. The individual cost components used to calculate FCOP are summarized in Table X.

$$FCOP = C_{labor} + C_{supervision} + C_{direct\ overhead} + C_{maintenance} + C_{plant\ overhead} + C_{tax\ \&\ insurance}$$

Equation 8: *Fixed Costs of Production*

The resulting total fixed cost of production was estimated to be $44.32 MM per year, as shown in the table below.

Table 11***:*** *Fixed Cost of Production Breakdown*		
Cost Component	**Basis**	**Annual Cost ($MM/yr)**
Operating Labor	15 operators × $70k/yr	1.05
Supervision	25% of labor	0.26
Direct Overhead	45% of labor + supervision	0.59
Maintenance	3% of ISBL	16.81
Plant Overhead	65% of labor + maintenance	12.16
Taxes and Insurance	2% of fixed capital	13.45
Total FCOP	—	44.32

5.2.2 Cash Cost of Production (CCOP)

The cash cost of production (CCOP) represents the total annual operating cost required to operate the facility. CCOP is calculated as the sum of the variable cost of production (VCOP) and the fixed cost of production (FCOP).

$$CCOP = VCOP + FCOP$$

Equation 9: *Cash Cost of Production*

Using the previously calculated operating costs:

$$VCOP = 243.8\ \$MM/yr,\ \ FCOP = 44.32\ \$MM/yr$$

the resulting cash cost of production becomes

$$CCOP = 243.8 + 44.32 = 288.1\ \$MM/yr$$

This value represents the total annual cost required to operate the plant and produce the final fuel products. The CCOP value will be used in the following section to evaluate plant profitability and determine the discounted cash flow rate of return (DCFROR). The operating cost breakdown is shown in the figure below.

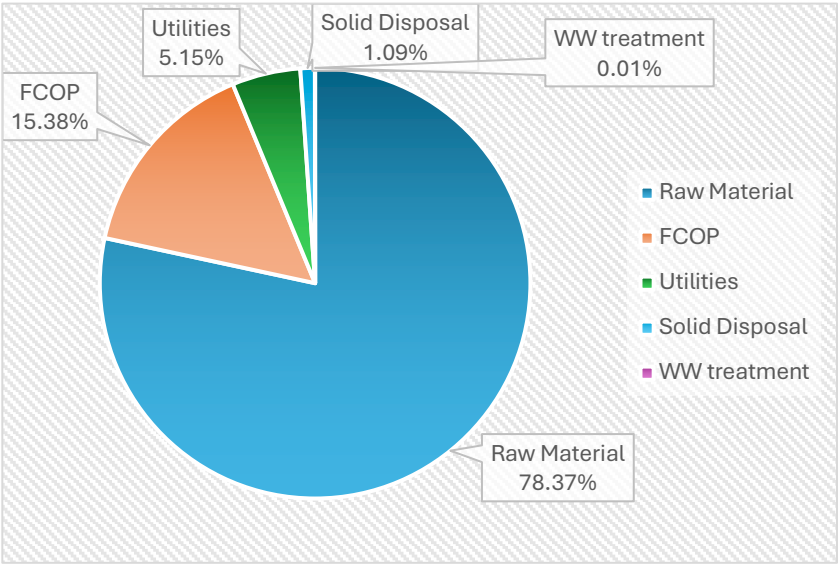


Figure 11: Operating Cost Breakdown (%)

5.3 Revenue and Gross Profit

Revenue for the proposed process was calculated from the sale of sustainable aviation fuel (SAF) and naphtha, together with the associated fuel credits. Market prices for the liquid fuel products were obtained from publicly available energy market data sources referenced in the bibliography. In addition to direct product sales, the process qualifies for fuel credits based on the carbon intensity of the produced fuels.

Because the process utilizes green hydrogen, the associated lifecycle CO_2 emissions are expected to be relatively low. As a result, the fuels produced by this process qualify for the higher end of available fuel credit values. Revenue was therefore calculated as the sum of product sales revenue and fuel credit revenue, as summarized in the following tables. Fuel prices

are taken from *Intratec Solutions* and *Airlines for America*, while credits are taken from *Aegis hedging*.

Table 12: *Annual Product Revenue*			
Product	**Production Rate (gal/hr)**	**Price ($/gal)**	**Annual Revenue ($MM/yr)**
SAF	17,680	2.5	353.6
Naphtha	6,028	1.6	77.16
Total Product Revenue	—	—	**430.76**

Table 13*: Annual Fuel Credit Revenue*		
Product	**Price ($/gal)**	**Annual Revenue ($MM/yr)**
SAF	1.4	198.0
Naphtha	2.45	118.15
Total Credit Revenue	—	**316.16**

The sum of product revenue and fuel credits gives the total annual revenue of the process, which amounted to 746.9 $MM/yr.

The gross profit of the process was determined by subtracting the cash cost of production (CCOP) from the total revenue. The results of this calculation are summarized in the table below.

Table 14*: Profit Summary*	
Category	**Value ($MM/yr)**
Total Revenue	746.92
Cash Cost of Production (CCOP)	288.1
Gross Profit	458.8

This gross profit value represents the annual earnings before depreciation, taxes, and financing considerations. These values are used in the following section to evaluate the economic

feasibility of the process through a discounted cash flow rate of return (DCFROR) analysis. The breakdown of the profits is shown in the pie chart below.

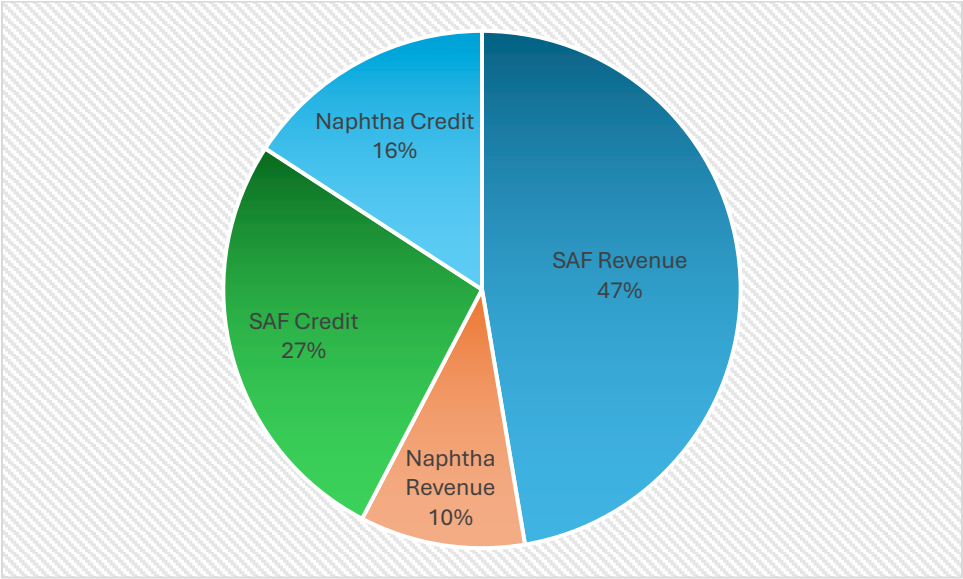


***Figure** 12: Revenue Breakdown*

5.4 Economic Feasibility and Discounted Cash Flow Analysis

The economic feasibility of the proposed process was evaluated using a discounted cash flow rate of return (DCFROR) analysis. The plant was assumed to have a 20-year operating lifetime, with an additional two years required for construction. During the construction period (years 1–2), the capital investment is distributed evenly and no revenue is generated. The plant is assumed to operate at 50% capacity during the first year of operation (year 3) before reaching full capacity for the remainder of the project lifetime.

As an initial measure of profitability, the simple payback period (PBP) and return on investment (ROI) were calculated using the total capital cost and annual gross profit determined in the previous sections. The simple payback period represents the time required for the project to recover the initial capital investment and is defined as

$$PBP = \frac{Total\ Capital\ Cost}{Gross\ Profit}$$

Equation 10: *Pay Back Period*

Using the calculated total capital cost and gross profit, the resulting payback period is approximately **2.14 years**. The return on investment (ROI) is calculated as

$$ROI = \frac{Gross\ Profit}{Total\ Capital\ Cost} \times 100$$

Equation 11: *Return on Investment*

which yields an ROI of approximately **46.8%**, indicating favorable profitability for the proposed process under the economic conditions considered.

A more rigorous evaluation was then performed using a discounted cash flow analysis, which accounts for the time value of money. The net present value (NPV) of the project is calculated as

$$NPV = \sum_{n=1}^{N} \frac{CF_n}{(1+i)^n}$$

Equation 12: *Net Present Value*

where CF_n represents the annual cash flow in year n, i is the discount rate, and N is the project lifetime. A discount rate of 15% was selected to represent the opportunity cost of capital and the investment risk associated with large-scale chemical process facilities.

Annual pre-tax cash flow was calculated using

$Pre\text{-}Tax$ Cash Flow = Revenue - VCOP - FCOP

Taxes were applied to the taxable income, defined as the difference between the pre-tax cash flow and the depreciation of capital equipment:

$$Taxable \text{ Income} = Pre\text{-}Tax \text{ Cash Flow - Depreciation}$$

The tax payment was then determined using

$$Tax = (Taxable\ Income)(Tax\ Rate)$$

An effective tax rate of 26.5% was used for the analysis. This value represents the combined impact of the U.S. federal corporate tax rate (21%) and a typical Louisiana state corporate tax contribution, which is appropriate for a chemical facility operating in the Gulf Coast region.

Depreciation of capital equipment was calculated using the Modified Accelerated Cost Recovery System (MACRS). Both MACRS-5 and MACRS-7 depreciation schedules were evaluated. Depreciation was applied only to the capital investment excluding working capital, since working capital is recovered at the end of the project lifetime.

The DCFROR is defined as the interest rate at which the net present value of the project equals zero. This value was determined in Excel using the Goal Seek function by varying the interest rate until the final NPV became zero. The resulting DCFROR values are summarized below.

***Table** 15: DCFOR Summary*

Depreciation Method	**DCFROR**
MACRS-5	**32.65%**
MACRS-7	**32.26%**

Both depreciation methods produce high rates of return, indicating strong economic performance for the proposed plant. The MACRS-5 depreciation schedule results in a slightly higher return because a greater fraction of the capital investment is recovered earlier in the plant lifetime.

The DCFROR values are primarily supported by the combination of product revenue and fuel credit revenue, which together provide sufficiently strong annual cash flow over the plant lifetime. The NPV curve in Figure 13 shows that the project remains negative during construction and startup years, then increases steadily after the plant reaches full operation, ultimately reaching a strongly positive value by the end of the project lifetime.

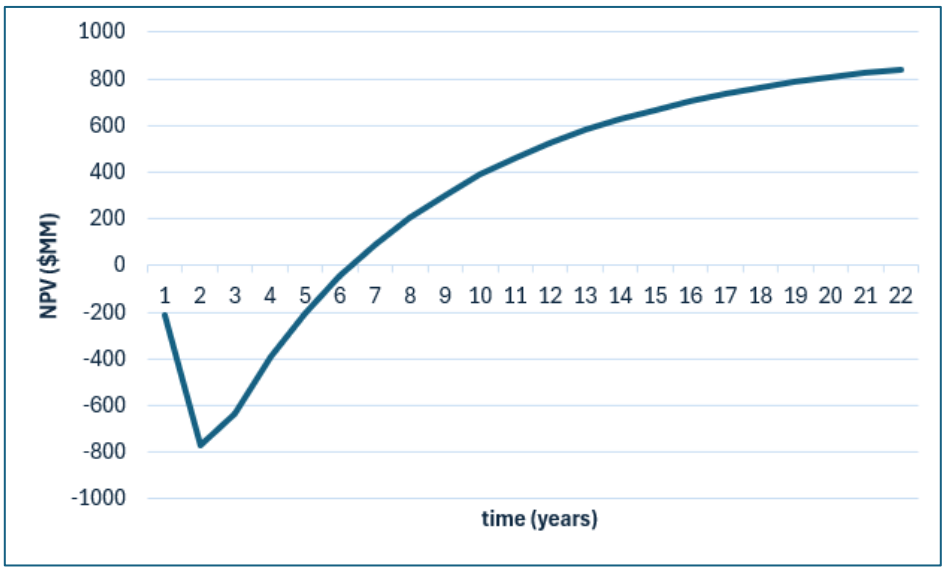


***Figure** 13: 7-year NPV analysis*

A test case was also performed where the assumption was made that no credits are received for the process. In this case, the DCFROR decreases to approximately **9.4%**. The NPV remains negative throughout the full project lifetime, as shown in Figure 14, although the curve increases steadily after startup. This result indicates that the process remains capable of generating positive annual cash flow, but the return is significantly weakened in the absence of fuel credits.

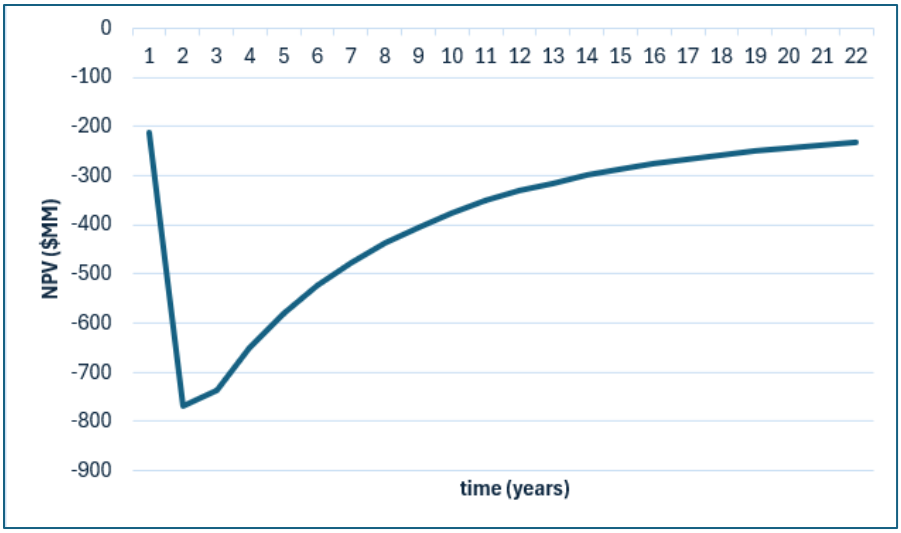


***Figure** 14: No Tax Credits NPV analysis*

Comparison of the two cases shows that fuel credits have a major effect on overall profitability. In the base case, the project generates a strongly positive NPV and a DCFROR above 30%, while the no-credit case produces a much lower rate of return and does not recover the discounted investment over the evaluation period. This demonstrates that policy incentives play an important role in improving the economic attractiveness of the proposed biomass-to-SAF process. Detailed year-by-year calculations used to determine pre-tax cash flow, depreciation, taxable income, tax payments, discount factors, and NPV are provided in the Appendix.

5.5 Sensitivity Analysis

A sensitivity analysis was conducted to evaluate how variations in key economic parameters affect the overall profitability of the proposed process. The parameters analyzed include installed capital cost (ISBL), variable operating cost (VCOP), fixed operating cost (FCOP), total revenue, and product revenue. Each parameter varied between -30% and +30% of its base value, and the resulting net present value (NPV) was calculated using both the MACRS-5 and MACRS-7 depreciation methods. The resulting trends are shown in Figures 15 and 16, while the no-credit sensitivity case is shown in Figure 17. Detailed numerical calculations are provided in the Appendix.

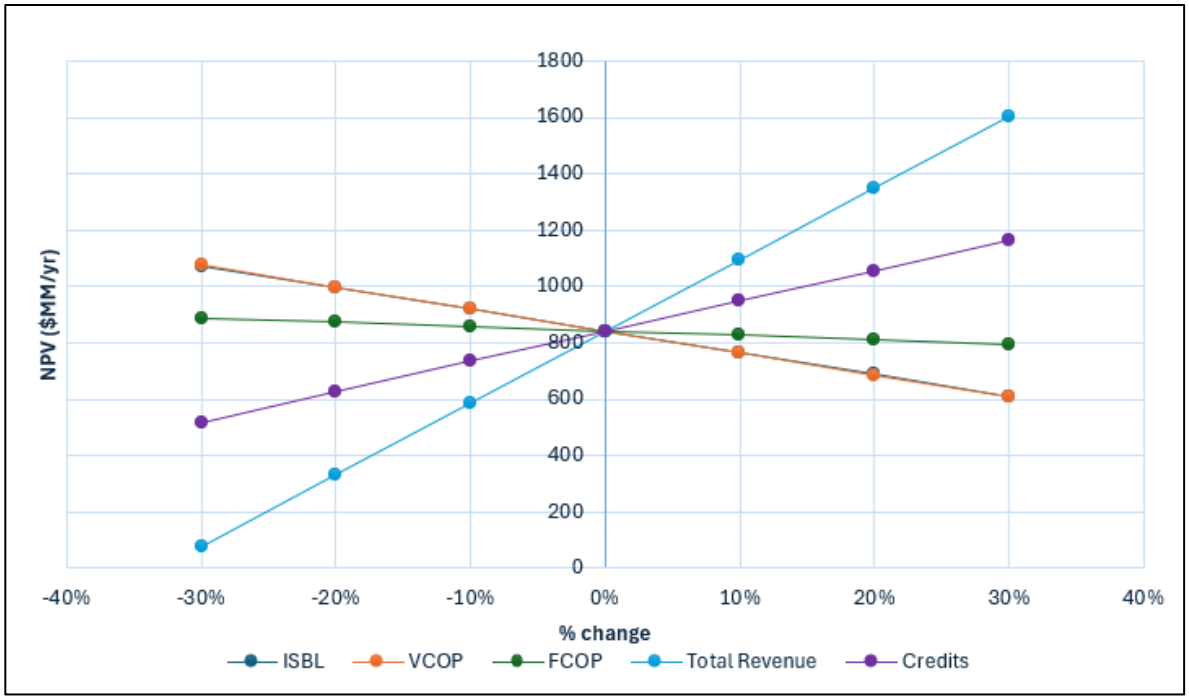


***Figure** 15: MACRS-7-year sensitivity analysis*

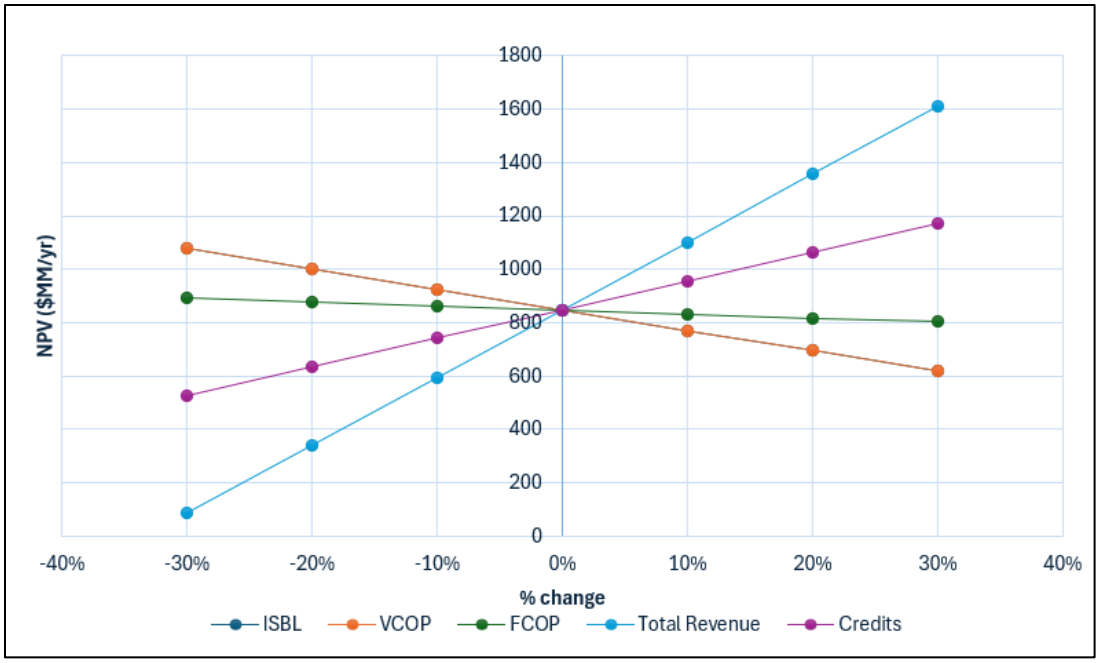


***Figure** 16: MACRS-5 year sensitivity analysis*



***Figure** 17: No-credit 7-year sensitivity analysis*

The sensitivity analysis shows that product revenue has the strongest influence on the economic performance of the process. In practical terms, this means that the project is highly dependent on the market value of its fuel products, particularly SAF. If SAF prices increase, the NPV improves significantly, while lower fuel prices quickly reduce profitability. This reflects the strong dependence of fuel production processes on commodity pricing.

Installed capital cost and variable operating cost have nearly identical effects on NPV. Changes in equipment cost or installation (ISBL) shift the NPV to a similar extent as changes in raw material and utility costs (VCOP), indicating that both upfront investment and recurring operating expenses are important economic drivers. Increasing either capital cost or operating cost reduces profitability, while reductions in these costs improve project performance. The overlap of these trends suggests that neither dominates, and both must be controlled to maintain favorable economics.

Fixed operating cost has the smallest effect on NPV. Changes in labor, maintenance, and overhead produce only minor variations in profitability compared to the other parameters. This indicates that fixed costs represent a relatively small portion of total annual expenses and are less critical in determining overall economic feasibility.

The sensitivity trends for MACRS-5 and MACRS-7 are nearly identical. This shows that the choice of depreciation method has little impact on the overall economic behavior of the plant. While depreciation affects the timing of tax benefits, it does not significantly influence the key drivers of profitability.

The no-credit case highlights the importance of fuel incentives. Without credits, the base-case NPV becomes negative, indicating that the project is not economically viable under

standard market conditions. Even with favorable changes in capital or operating costs, the process struggles to achieve positive returns, while improvements in fuel prices are required to approach profitability.

Overall, the results indicate that the economic feasibility of the plant is most sensitive to fuel prices, with capital cost and operating cost having comparable secondary effects. Fixed costs and depreciation method play a smaller role. Most importantly, the process is highly dependent on fuel credits, which are essential for maintaining positive project economics and enabling commercial viability.

6. Appendix

Appendix I: Design Basis

This design is based on a ~10,000 barrels per day (bbl/day) sustainable aviation fuel (SAF) production facility using biomass gasification, syngas conditioning, low-temperature cobalt Fischer–Tropsch (FT) synthesis, and downstream hydrocracking/hydrotreating. The overall configuration and scale are benchmarked against commercial FT-SAF projects under development along the U.S. Gulf Coast—notably the DG Fuels plant in Louisiana, which will employ Johnson Matthey and bp's FT CANS™ technology at a capacity of 13,000 bbl/day. This provides a realistic industrial reference point for sizing, performance expectations, and catalyst inventory at the scale considered in this study.

St. James Parish, Louisiana is selected as the plant location due to its favorable combination of biomass supply, industrial infrastructure, hydrogen availability, and proximity to emerging FT-SAF developments. Louisiana ranks among the top U.S. producers of forestry residues, logging waste, and wood-processing by-products, which form the primary biomass

feedstock. The Mississippi River petrochemical corridor offers extensive utilities, existing pipelines, and refinery-adjacent infrastructure that reduce capital and operating costs for a biomass-to-liquids (BtL) facility. The region also benefits from strong hydrogen production capacity supported by ammonia and petrochemical operations, as well as growing SAF demand and supportive federal incentives such as the Inflation Reduction Act's SAF tax credits. Deep-water access through the ports of Baton Rouge and New Orleans simplifies logistics for catalyst delivery, equipment transport, and export of finished SAF.

To meet the target of 10,000 bbl/day SAF, the FT synthesis and upgrading section is designed for approximately 1,450 tonnes/day of total FT liquids, of which about 1,250 tonnes/day fall within the SAF carbon range (C_8–C_{16}), with the remainder as naphtha co-products. This production scale is consistent with biomass-to-liquids studies reporting overall biomass-to-liquids efficiencies of ~40% (LHV basis) for gasification followed by FT synthesis and upgrading.

At the front end of the process, the plant processes 5,700 tonnes/day of woody biomass, sourced from regional forestry residues and mill by-products. The biomass is dried on-site and fed to the gasifier to produce a raw syngas containing CO, H_2, CO_2, steam, and light hydrocarbons.

Biomass gasification naturally produces syngas with a low H_2/CO ratio, which is not suitable for low-temperature cobalt FT catalysts and FT synthesis in general. Because cobalt FT operation requires a target H_2/CO ratio near 2, the design incorporates a water–gas shift (WGS) reactor to partially convert CO and steam to H_2 and CO_2. Typical design studies report steam-to-syngas ratios between 0.4 and 0.7 kg steam per kg of dry syngas; here a value of 0.5 kg steam

per kg of conditioned syngas is selected, resulting in a steam utility load of approximately 1,000 tonnes/day for syngas conditioning at the required FT reactor feed rate.

The FT synthesis section uses a low-temperature cobalt catalyst, selected for its high activity and strong selectivity toward long-chain linear paraffins, producing heavy wax (C_{20}^{+}) and middle distillate (C_9–C_{20}) hydrocarbons. Catalyst inventory is scaled from published industrial data indicating that a 100,000 bbl/day FT facility requires roughly 646 tonnes of cobalt. Normalizing to the capacity of this design yields an estimated 65 tonnes of cobalt, corresponding to approximately 320 tonnes of supported catalyst assuming a 20 wt% Co formulation.

Downstream of FT synthesis, the heavy wax and middle-distillate fractions are routed to hydrocracking and hydrotreating units to produce the desired blend of SAF and naphtha. The detailed hydrogen consumption associated with these upgrading steps is addressed separately in the mass balance section of the report.

Appendix II: Block Flow Diagram

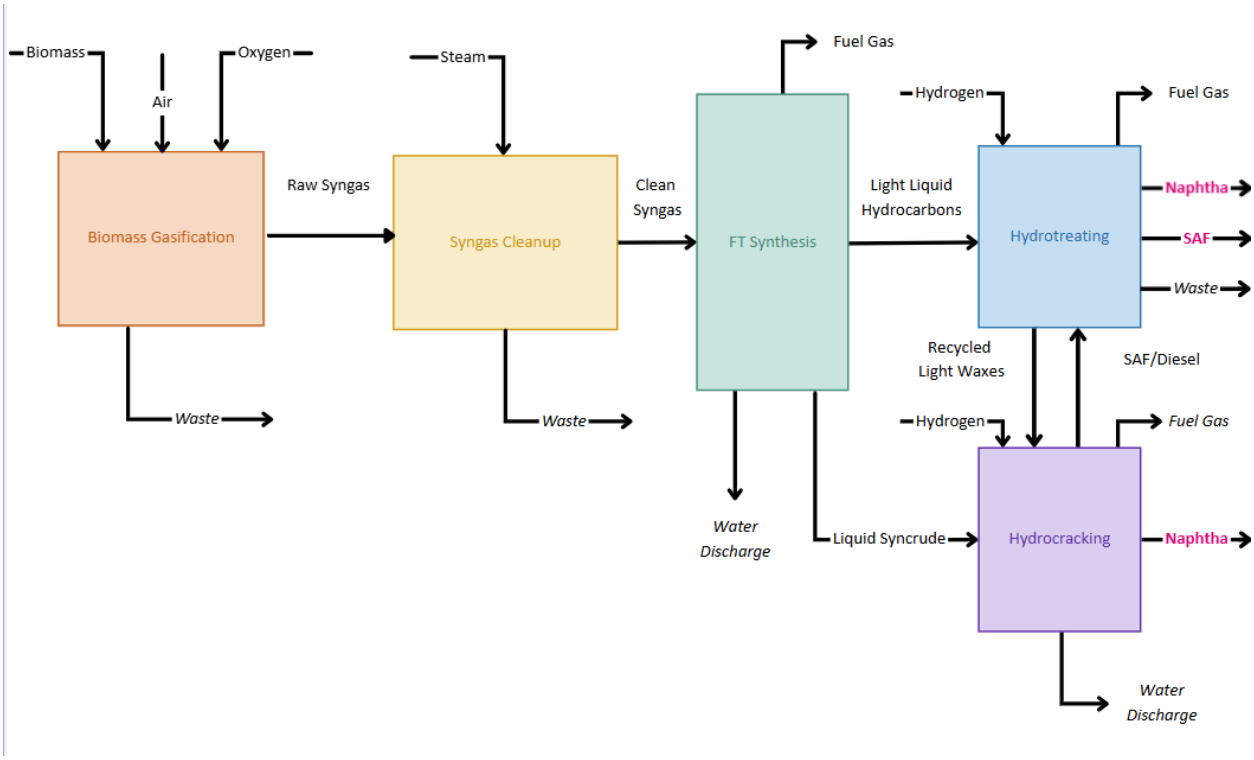


***Figure** 18: Final Block Flow Diagram*

Appendix III: Process Flow Diagram and Material balance

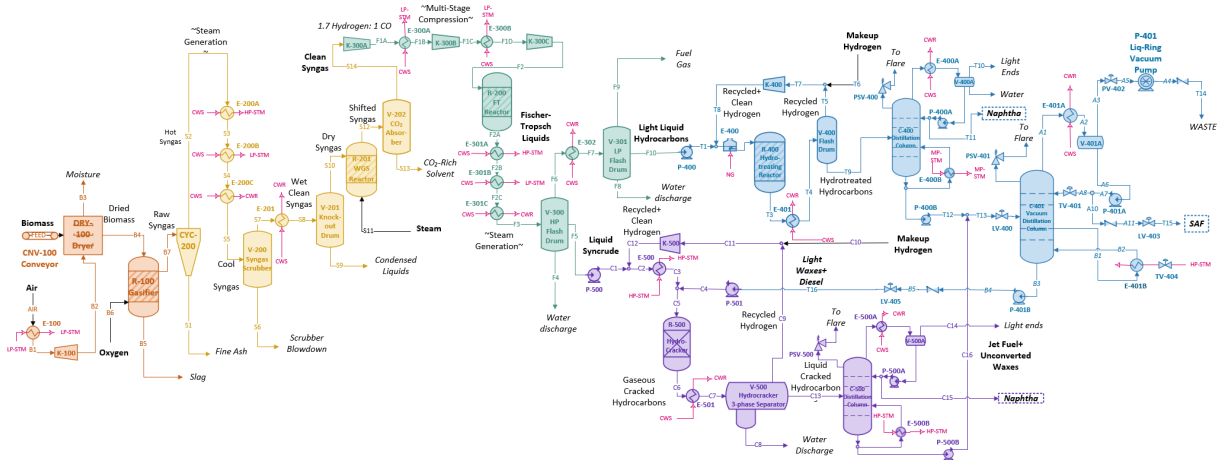


***Figure** 19: Final Process Flow Diagram*

The overall material balance is based on the final Aspen Plus v14 simulation and is organized by the major blocks in the flowsheet: Biomass Gasification (100), Syngas Cleanup/Conditioning (200), Fischer–Tropsch Synthesis (300), Hydrotreating (400), and Hydrocracking (500). The final configuration reflects the process changes made during optimization, most importantly the reduced biomass basis and the decision to eliminate diesel as an external co-product by routing diesel-range material and heavy wax back into the upgrading loop, so that they are ultimately converted into jet-range material.

The front-end biomass feed enters the process at approximately 524,000 lb/hr of dry woody biomass. After drying and the introduction of the oxidant, the gasifier produces a raw syngas stream B7 with a total flow rate of approximately 336,000 lb/hr. This stream contains approximately 262,000 lb/hr CO, 24,000 lb/hr H_2, 17,500 lb/hr CO_2, and 20,000 lb/hr CH_4, along with water vapor, minor light hydrocarbons, nitrogen, and approximately 5,000 lb/hr of ash. The cyclone removes the solid ash as stream S1, reducing the vapor-phase syngas flow to approximately 331,000 lb/hr in stream S2. No significant hydrocarbon loss occurs in this solids separation step, and carbon is conserved within the gas phase.

The syngas cleanup section conditions the gas for catalytic synthesis. After cooling, scrubbing, and condensation, water is partially removed in liquid knockout streams (S6 and S8), reducing the water loading prior to shift conversion. The water–gas shift reactor increases hydrogen concentration by converting CO, and the downstream CO2 removal unit separates CO2 into stream S13. Following these steps, the lean syngas stream S14 has a total mass flow of approximately 245,000 lb/hr. This stream contains approximately 218,000 lb/hr CO and 27,000 lb/hr H2, with CO2 reduced to negligible levels and no remaining solids. The reduction in total

mass from the raw syngas basis reflects the removal of CO2, condensed water, and solids. The S14 stream represents the total fresh syngas feed to the FT reactor before recycle blending.

In the Fischer–Tropsch section, the reactor feed F1 equals the conditioned syngas flow of approximately 245,000 lb/hr. Across the reactor, total mass is conserved; however, the species distribution shifts substantially. The reactor effluent F2/F3 includes approximately 33,000 lb/hr of reaction water formed, 34,000 lb/hr CO2 generated or carried through, and multiple hydrocarbon fractions. Hydrocarbon products are represented in lumped carbon-number ranges: C5–C7 (~15,000 lb/hr), C8–C16 (~47,000 lb/hr), C17–C22 (~38,500 lb/hr), and C23–C40 (~23,975 lb/hr), with trace C42–C60 components. High- and low-pressure flash separation splits this effluent into a hydrocarbon liquid stream, F5 (approximately 119,000 lb/hr), and a gas stream, F6 (approximately 126,000 lb/hr). The gas stream contains unreacted CO and H2, light hydrocarbons, CO2, and water vapor and is partially recycled to maintain reactor utilization, with a controlled purge to prevent inert buildup. The liquid stream F5 contains the majority of the C8+ material and serves as the feed to the upgrading unit.

Hydrotreating receives the FT liquid stream F10, which represents the stabilized portion of F5 after initial separation, with a flow rate of approximately 19,000 lb/hr, as shown in the detailed stream table. Hydrogen addition and mild conversion redistribute minor quantities of light gases and water but do not significantly alter total hydrocarbon mass. Across the hydrotreating reactor and flash separation sequence (T1–T15), total hydrocarbon mass is conserved within expected hydrogen addition and purge limits. The stabilized liquid leaving the hydrotreating block is then fractionated to separate lighter naphtha-range components from heavier material directed to hydrocracking.

The hydrocracking section processes the heavy wax fraction originating from the FT separation (C23–C40 and C17–C22 material) along with heavier hydrotreated streams. The wax feed case represented by stream F6 entering the hydrocracking block shows approximately 110,000 lb/hr of heavy hydrocarbons prior to hydrogen addition. After mixing with hydrogen (C2/C3), the total reactor feed increases to approximately 118,000 lb/hr due to hydrogen incorporation. Within the hydrocracker, C20+ components are cracked and isomerized into lighter fractions, redistributing mass into C5–C7, C8–C16, and C17–C22 ranges while generating a light gas fraction. Reactor outlet separation removes a vapor-phase stream containing C1–C4 and light components, while the liquid phase proceeds to fractionation. Unconverted heavy material is internally recycled within the hydrocracking loop to increase overall conversion of wax into jet-range hydrocarbons.

When hydrotreated and hydrocracked liquid streams are combined, the total liquid production aligns with approximately 1,450 tonnes/day of finished fuels. Of this total, approximately 1,250 tonnes/day are recovered in the C8–C16 SAF range, with the balance primarily in the naphtha range. Diesel-range hydrocarbons are not withdrawn as a product stream in the final configuration but are internally recycled to the hydrocracking section. Overall mass closure across the entire flowsheet is maintained within Aspen's numerical tolerance, and the stream data confirm conservation of total mass through each major block, with expected redistribution among the vapor, liquid, and solid phases. The mass balances tables and their respective process flow diagram sections from earlier in the report are shown below.

Biomass Gasification

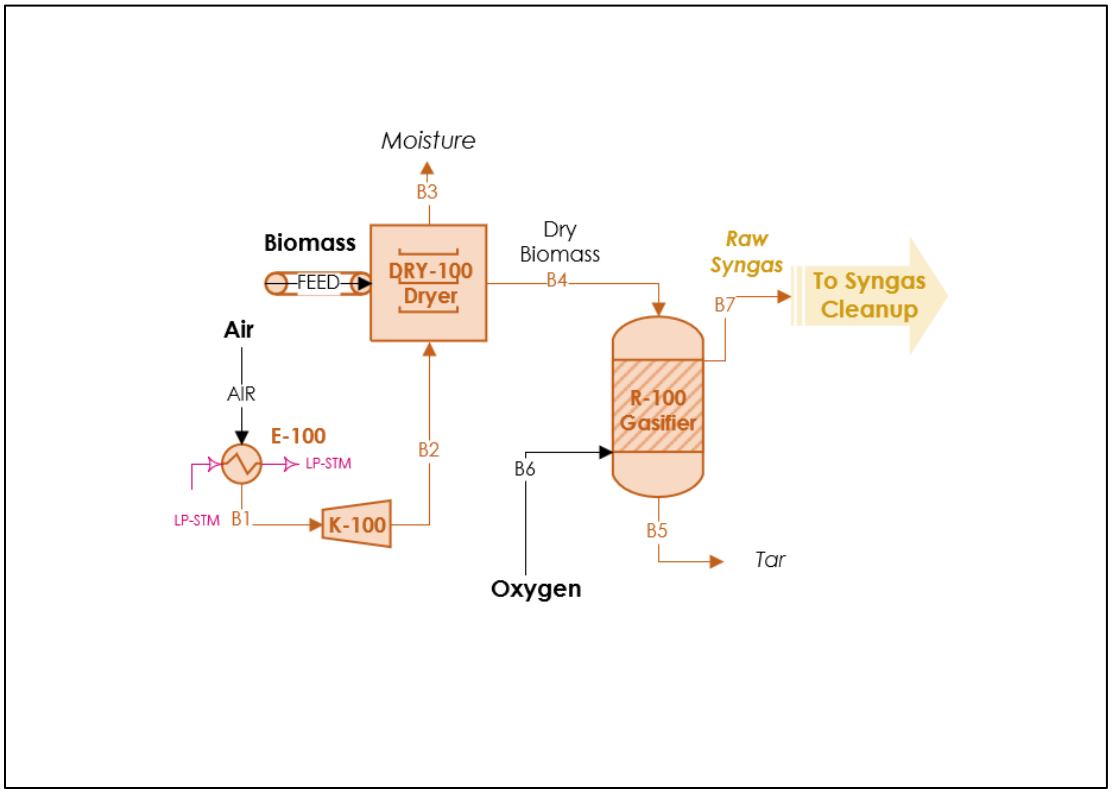


Table 17: Mass Balance - Biomass gasifcations Section-100										
Component	Units	Feed	Air	B1	B2	B3	B4	B5	B6	B7
Phase	Unitless	Solid	Vapor	Vapor	Vapor	Vapor	Solid	Solid	Vapor	Vapor
Temperature	oF	77	77	309	140	250	250	1472	77	1,397
Pressure	psia	14.7	19.7	48.7	43.7	14.7	14.7	15	15	15
Biomass	lb/hr	523,598	-	-	-	-	327,249	-	-	-
CO	lb/hr	-	-	-	-	-	-	-	-	261,959
H_2	lb/hr	-	-	-	-	-	-	-	-	23,625
H_2O	lb/hr	-	-	-	-	196,349	-	-	-	7,003
CO_2	lb/hr	-	-	-	-	-	-	-	-	17,473
CH_4	lb/hr	-	-		-	-	-	-	-	20,251
C_2H_6	lb/hr	-	-	-	-	-	-	-	-	1
C_3H_8	lb/hr	-	-	-	-	-	-	-	-	trace
O_2	lb/hr	-	-	-	-	-	-	-	11,023	trace
N_2	lb/hr	-	-	-	-	-	-	-	-	393
Ash	lb/hr	-	-	-	-		-	-	-	5,044
Air	lb/hr	-	220	220	220	220	-	-	-	-
Tar	lb/hr	-	-	-	-	-	-	2,522	-	-
Total	**lb/hr**	**523,598**	**220**	**220**	**220**	**196,570**	**327,249**	**2,522**	**11,023**	**335,750**

Syngas Cleanup

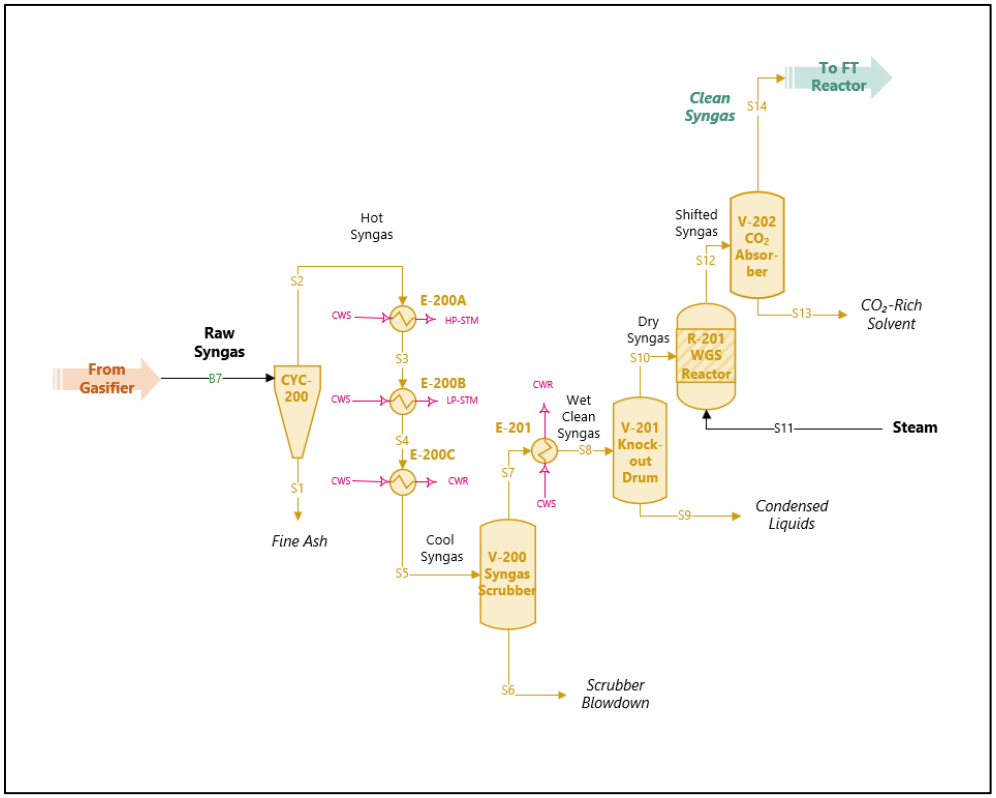


Table 18: Mass Balance - Syngas Section-200 Part 1								
Component	Units	B7	S1	S2	S3	S4	S5	S6
Phase	Unitless	Vapor	Solid	Vapor	Vapor	Vapor	Vap-Liq	Liquid
Temperature	°F	1,397	1,397	1,397	492	280	100	100
Pressure	psia	15	15	15	185	180	175	180
CO	lb/hr	261,959	-	261,959	261,959	261,959	261,959	trace
H_2	lb/hr	23,625	-	23,625	23,625	23,625	23,625	trace
H_2O	lb/hr	7,003	-	7,003	7,003	7,003	7,003	4,958
CO_2	lb/hr	17,473	-	17,473	17,473	17,473	17,473	trace
CH_4	lb/hr	20,251	-	20,251	20,251	20,251	20,251	trace
C_2H_6	lb/hr	1	-	1	1	1	1	trace
C_3H_8	lb/hr	trace	-	trace	trace	trace	trace	trace
O_2	lb/hr	trace	trace	trace	trace	trace	trace	trace
N_2	lb/hr	393	-	393	393	393	393	-
Ash	lb/hr	5,044	5,044	-	-	-	-	-
Total	**lb/hr**	**335,750**	**5,044**	**330,706**	**330,706**	**330,706**	**330,706**	**4,958**

Table 19: Mass Balance - Syngas Section-200 Part 2									
Component	Units	S7	S8	S9	S10	S11	S12	S13	S14
Phase	Unitless	Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor	Vapor
Temperature	°F	100	86	86	86	212	650	650	650
Pressure	psia	180	175	175	175	15	145	145	145
CO	lb/hr	261,959	261,959	trace	261,959	-	217,814	-	217,814
H_2	lb/hr	23,625	23,625	trace	23,625	-	26,802	-	26,802
H_2O	lb/hr	2,045	2,045	881	1,164	39,396	12,168	12,168	-
CO_2	lb/hr	17,473	17,473	trace	17,473	-	86,834	86,834	-
CH_4	lb/hr	20,251	20,251	trace	20,251	-	20,251	20,251	-
C_2H_6	lb/hr	1	1	trace	1	-	1	1	-
C_3H_8	lb/hr	trace	trace	trace	trace	-	trace	trace	-
O_2	lb/hr	trace	trace	trace	trace	trace	trace	trace	-
N_2	lb/hr	393	393	trace	393	-	393	393	-
Ash	lb/hr	-	-	-	-	-	-	-	-
Total	**lb/hr**	**325,747**	**325,747**	**881**	**324,866**	**39,396**	**364,263**	**119,647**	**244,616**

FT Synthesis

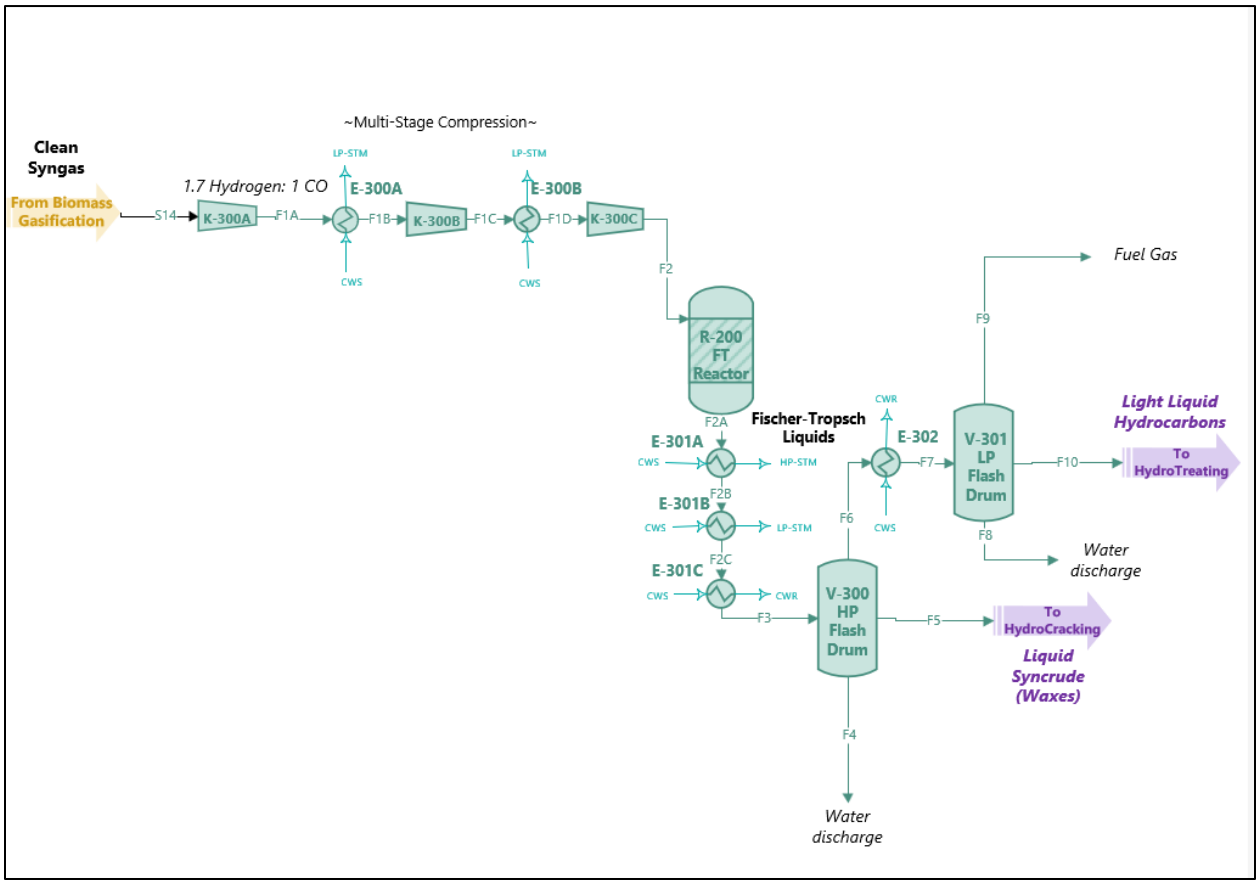


Table 20: Mass Balance - FT Section-300 Part 1									
Component	Units	S13	F1A	F1B	F1C	F1D	F2	F2A	F2B
Phase	Unitless	Vapor	Vapor	Vapor	Vapor	Vapor	Vap-Liq	Vap-Liq	Liquid
Temperature	°F	650	769	750	851	650	734	580	2492
Pressure	psia	145	195	190	240	235	290	290	285
H_2	lb/hr	26,802	26,802	26,802	26,802	26,802	26,802	17	17
CO	lb/hr	217,814	217,814	217,814	217,814	217,814	217,814	21,671	21,671
H_2O	lb/hr	-	-	-	-	-	-	33,047	33,047
CO_2	lb/hr	-	-	-	-	-	-	34,053	34,053
H_2S	lb/hr	-	-	-	-	-	-	trace	trace
C_1-C_4 (H^+)	lb/hr	-	-	-	-	-	-	31,818	31,818
C_1-C_4	lb/hr	-	-	-	-	-	-	trace	trace
C_5-C_7	lb/hr	-	-	-	-	-	-	14,774	14,774
C_8-$C16$	lb/hr	-	-	-	-	-	-	46,842	46,842
C_{17}-C_{22}	lb/hr	-	-	-	-	-	-	38,452	38,452
C_{23}-C_{40}	lb/hr	-	-	-	-	-	-	23,929	23,929
C_{42}-C_{60}	lb/hr	-	-	-	-	-	-	13	13
Total	**lb/hr**	**244,616**	**244,616**	**244,616**	**244,616**	**244,616**	**244,616**	**244,616**	**244,616**

Table 21: Mass Balance - FT Section-300 Part 2										
Component	Units	F2C	F3	F4	F5	F6	F7	F8	F9	F10
Phase	Unitless	Vap-Liq	Vap-Liq	Liquid	Liquid	Vapor	Vap-Liq	Liquid	Vapor	Liquid
Temperature	°F	280	257	296	296	296	80	122	122	122
Pressure	psia	280	275	203	203	203	198	300	300	300
H_2	lb/hr	17	17	-	trace	17	17	trace	17	trace
CO	lb/hr	21,671	21,671	-	114	21,557	21,557	trace	21,469	87
H_2O	lb/hr	33,047	33,047	8,862	2,032	22,153	22,153	21,752	278	122
CO_2	lb/hr	34,053	34,053	3	586	33,463	33,463	6	32,687	767
H_2S	lb/hr	trace	trace	trace	Trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	31,818	31,818	1	1,343	30,475	30,475	trace	27,518	2948
C_1-C_4	lb/hr	trace	trace	trace	Trace	trace	trace	trace	trace	trace
C_5-C_7	lb/hr	14,774	14,774	trace	4,101	10,673	10,673	trace	3,069	7589
C_8-$C16$	lb/hr	46,842	46,842	trace	39,431	7,411	7,411	trace	248	7119
C_{17}-C_{22}	lb/hr	38,452	38,452	trace	38,334	117	117	trace	trace	116
C_{23}-C_{40}	lb/hr	23,929	23,929	trace	23,922	7	7	trace	trace	7
C_{42}-C_{60}	lb/hr	13	13	trace	13	trace	trace	trace	trace	trace
Total	**lb/hr**	**244,616**	**244,616**	**8,867**	**119,877**	**125,872**	**125,872**	**21,758**	**85,286**	**18754**

Hydrotreating

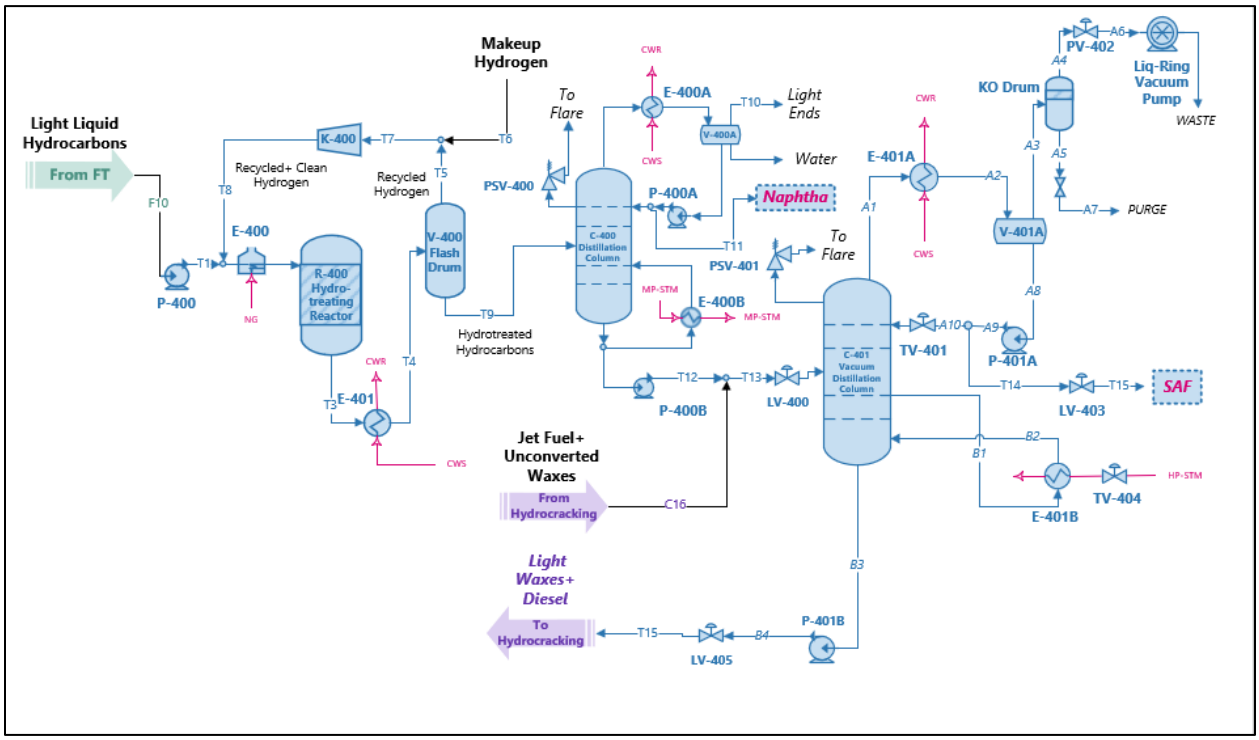


Table 22: Mass Balance - Hydrotreating Section-400 Part 1										
Component	Units	F10	T1	T2	T3	T4	T5	T6	T7	T8
Phase	Unitless	Liquid	Liquid	Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor
Temperature	°F	122	131	572	608	122	122	77	101	445
Pressure	psia	300	865	860	865	860	354	100	100	900
H_2	lb/hr	trace	trace	trace	trace	trace	trace	2	5	T5
CO	lb/hr	87	87	87	162	162	75	-	75	75
H_2O	lb/hr	122	122	122	123	123	1	-	1	1
CO_2	lb/hr	767	767	767	882	882	115	-	115	115
H_2S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	2,948	2,948	2,948	3,045	3,045	97	-	97	97
C_1-C_4	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_5-C_7	lb/hr	7,589	7,589	7,589	7,600	7,600	11	-	11	11
C_8-C_{16}	lb/hr	7,119	7,119	7,119	7,120	7,120	1	-	1	1
C_{17}-C_{22}	lb/hr	116	116	116	116	116	trace	-	trace	trace
C_{23}-C_{44}	lb/hr	7	7	7	7	7	trace	-	trace	trace
Total	**lb/hr**	**18,754**	**18,754**	**18,754**	**19,054**	**19,054**	**305**	**2**	**305**	**305**

Table 23: Mass Balance - Hydrotreating Section-400 Part 2									
Component	Units	T9	T10	T11	T12	C16	T13	T14	T15
Phase	Unitless	Liquid	Vapor	Liquid	Liquid	Liquid	Vap-Liq	Liquid	Liquid
Temperature	^{o}F	122	91	91	323	437	420	149	443
Pressure	psia	354	45	45	45.2	45.2	45.2	1.1	1.3
H_2	lb/hr	2	2	trace	trace	trace	trace	trace	trace
CO	lb/hr	87	87	1	trace	-	trace	trace	trace
H_2O	lb/hr	122	48	72	2	trace	2	2	trace
CO_2	lb/hr	767	737	29	trace	trace	trace	trace	trace
H_2S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	2,948	1,990	940	17	-	17	17	trace
C_1-C_4	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace
C_5-C_7	lb/hr	7,589	466	4,617	2,505	2,239	4,744	4,744	trace
C_8-C_{16}	lb/hr	7,119	trace	12	7,106	82,553	89,659	89,467	193
C_{17}-C_{22}	lb/hr	116	trace	trace	116	3,711	3,826	930	2,896
C_{23}-C_{44}	lb/hr	7	trace	trace	7	100	107	-	107
Total	**lb/hr**	**18,756**	**3,331**	**5,672**	**9,753**	**88,603**	**98,356**	**95,160**	**3,196**

Hydrocracking

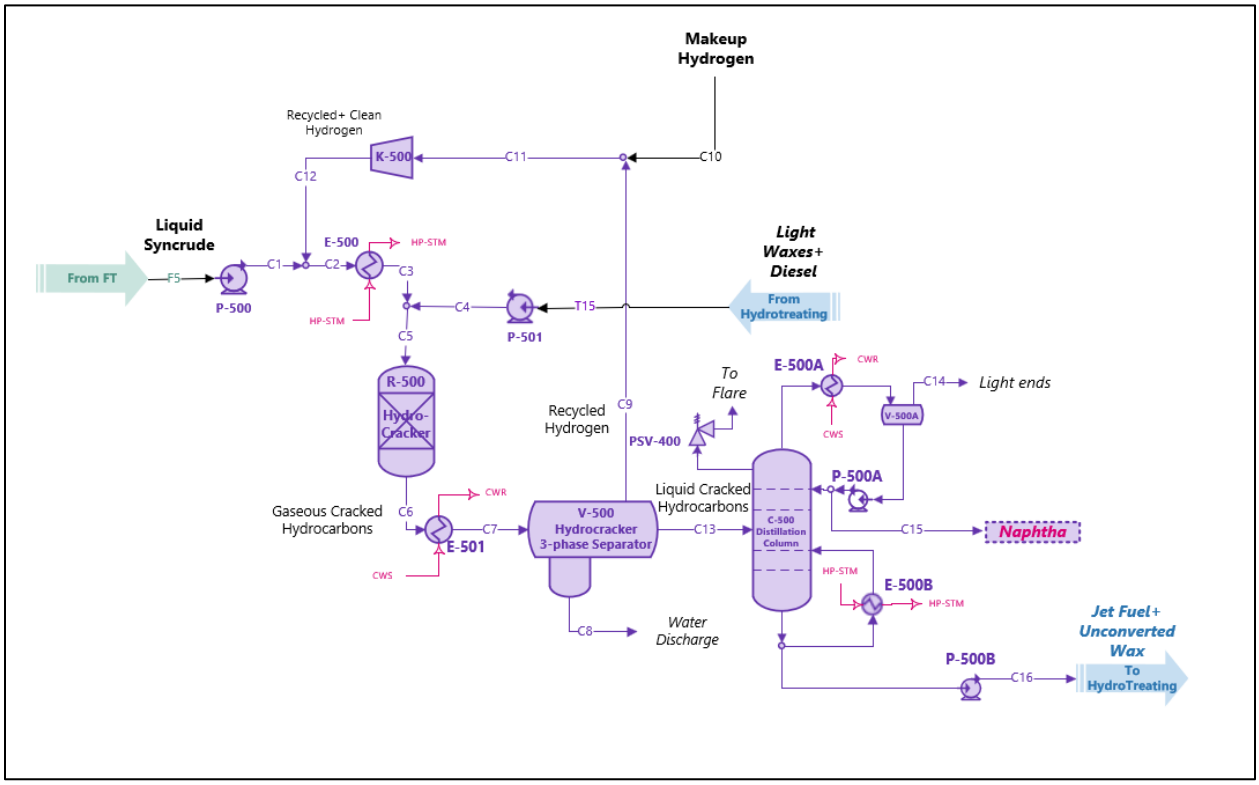


Table 24: Mass Balance - Hydrocracking Section-500 Part 1										
Component	Units	F6	C1	C2	C3	T15	C4	C5	C6	C7
Phase	Unitless	Liquid	Liquid	Vap-Liq	Vap-Liq	Liquid	Liquid	Vap-Liq	Liquid	Liquid
Temperature	°F	296	301	298	350	443	449	352	350	122
Pressure	psia	203	1,000	1,000	995	1	1,000	995	915	910
H_2	lb/hr	trace	trace	4,600	4,600	trace	trace	4,600	50	50
CO	lb/hr	115	115	115	115	trace	trace	115	trace	trace
H_2O	lb/hr	2,030	2,030	2,031	2,031	trace	trace	2,031	2,242	2,242
CO_2	lb/hr	591	591	592	592	trace	trace	592	50	50
H_2S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	1,354	1,354	1,354	1,354	trace	trace	1,354	trace	trace
C_1-C_4	lb/hr	trace	trace	1	1	trace	trace	1	100	100
C_5-C_7	lb/hr	4,135	4,135	4,140	4,140	trace	trace	4,140	29,119	29,119
C_8-C_{16}	lb/hr	39,566	39,566	39,566	39,566	193	193	39,759	82,613	82,613
C_{17}-C_{22}	lb/hr	38,410	38,410	38,410	38,410	2,896	2,896	41,306	3,711	3,711
C_{23}-C_{40}	lb/hr	23,968	23,968	23,968	23,968	107	107	24,075	100	100
C_{42}-C_{60}	lb/hr	13	13	13	13	trace	trace	13	trace	trace
Total	**lb/hr**	**110,182**	**110,182**	**114,789**	**114,789**	**3,196**	**3,196**	**117,985**	**117,985**	**117,985**

Table 25: Mass Balance - Hydrocracking Section-500 Part 2										
Component	Units	C8	C9	C10	C11	C12	C13	C14	C15	C16
Phase	Unitless	Liquid	Vapor	Vapor	Vapor	Vapor	Liquid	Vapor	Liquid	Liquid
Temperature	°F	122	122	77	77	354	122	138	138	437
Pressure	psia	300	300	300	300	1,000	300	45	45	45
H_2	lb/hr	trace	7	4,593	4,600	4,600	43	42	1	trace
CO	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
H_2O	lb/hr	1,752	trace	trace	trace	trace	489	73	417	trace
CO_2	lb/hr	trace	1	trace	1	1	49	40	9	trace
H_2S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4	lb/hr	trace	1	trace	1	1	100	48	52	trace
C_5-C_7	lb/hr	trace	5	trace	5	5	29,114	1,178	25,697	2,239
C_8-C_{16}	lb/hr	trace	trace	trace	trace	trace	82,613	trace	60	82,553
C_{17}-C_{22}	lb/hr	trace	trace	trace	trace	trace	3,711	trace	trace	3,711
C_{23}-C_{40}	lb/hr	trace	trace	trace	trace	trace	100	trace	trace	100
C_{42}-C_{60}	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
Total	**lb/hr**	**1,752**	**14**	**4,593**	**4,607**	**4,607**	**116,219**	**1,381**	**26,236**	**88,603**

Appendix IV: Equipment List

The majority of process equipment in this design is constructed from carbon steel due to its low cost, adequate mechanical strength, and suitability for handling non-corrosive hydrocarbon and gas streams. However, several units require more specialized materials due to elevated temperatures, hydrogen service, and potential corrosion. The water-gas shift reactor (R-201) and hydrotreater (R-400) are constructed from 1.25Cr–0.5Mo alloy steel, while the hydrocracker (R-500) uses 2.25Cr–1.0Mo alloy steel, as these chromium–molybdenum alloys provide improved resistance to high-temperature hydrogen attack (HTHA) and enhanced mechanical strength at elevated temperatures. The Fischer–Tropsch reactor (R-300) is constructed from 316 stainless steel, which offers superior resistance to corrosion, thermal cycling, and catalyst-related chemical exposure in slurry-phase operation. These material selections are consistent with industry practice for high-temperature, high-pressure hydrogenation and synthesis processes, where carbon steel alone is insufficient to ensure long-term equipment integrity. The following tables highlight the main equipment needed to run the process as well as the recommended material of construction.

Figure 20: *Biomass Gasification Section-100*		
Equipment	**Description**	**Material of Construction**
DRY-100	Biomass Drier	Carbon Steel
R-100	Entrained Flow Gasifier	Carbon Steel

Figure 21: *Syngas Cleanup Section-200*		
Equipment	**Description**	**Material of Construction**
CYC-200	Syngas Cyclone	Carbon Steel
V-200	Syngas Scrubber	Carbon Steel
V-201	Knock-Out Drum	Carbon Steel

R-201	Water Gas Shift Reactor (Packed Bed Reactor)	1.25Cr-0.5Mo Alloy
V-202	CO_2 Scrubber	Carbon Steel

Figure 22: *FT Reactor Section-300*		
Equipment	**Description**	**Material of construction**
R-300	FT Reactor (Slurry Bubble Column Reactor)	316 S.S
V-300	HP Flash Drum	Carbon Steel
V-301	LP Flash Drum	Carbon Steel

Figure 23: *Hydrotreating Section-400*		
Equipment	**Description**	**Material of construction**
R-400	Hydrotreater Reactor (Fixed-Bed Hydrogenation Reactor)	1.25Cr-0.5Mo Alloy
V-400	Flash Drum	Carbon Steel
C-400	Tray Distillation Column (Sieve Tray)	Carbon Steel
C-401	Packed Vacuum Distillation Column	Carbon Steel
P-400	Centrifugal Pump	Carbon Steel
P-400A	Centrifugal Pump	Carbon Steel
P-401A	Centrifugal Pump	Carbon Steel

Figure 24: *Hydrocracking Section*		
Equipment	**Description**	**Material of construction**
R-500	Hydrocracker Reactor (Fixed Bed Reactor)	2.255Cr-1.0 Mo Alloy
V-500	3-phase Separator	Carbon Steel
C-500	Tray Fractionation Column (Sieve Tray)	Carbon Steel
P-500	Centrifugal Pump	Carbon Steel
P-501	Centrifugal Pump	Carbon Steel
P-500A	Centrifugal Pump	Carbon Steel

Appendix V: Equipment Sizing

The following table displays the equipment specifications in order to meet the requirements of our plant's production.

Figure 25: *Heat Exchanger Duties*

Equipment	Duty (KW)
E-100	5
E-101	13300
E-102	900
E-200A	10500
E-200B	10300
E-201	900
E-300A	850
E-300B	8900
E-301A	5825
E-301B	9800
E-301C	1160
E-302	9890
E-400	1890
E-401	2070
E-500	1290
E-501	4670

Figure 26: *Pump Flow Requirements*

Unit	GPM
P-500	449.64
P-501	13.55
P-500 A	65.17
P-400	65.85
P-400A	12.23
P-401A	26.54

Figure 27: *Column Sizing*				
Distillation column	**Diameter (ft)**	**Height (ft)**	**R.R**	**D/F**
C-400	6	58	0.395	0.48
C-401	19.5	70	0.080	0.967
C-500	9	64	0.669	0.237

Appendix VI: Sample Hand Calculations

Hand calculations were performed to validate the Aspen RADFRAC design for the C-401 vacuum distillation column by estimating (1) number of stages, (2) column height, and (3) column diameter. Because the system is multicomponent (SAF/diesel-range hydrocarbons), the Fenske–Underwood–Gilliland (FUG) method was applied: the Fenske equation establishes a lower bound on the required stages at total reflux, the Underwood equations determine the minimum reflux condition consistent with the specified split, and the Gilliland correlation relates reflux to the practical number of stages required under operating conditions. The hand calculations were intended to provide a close comparison to Aspen trends rather than replicate a full rigorous stage-by-stage solution.

The vacuum distillation column (C-401) was modelled with a total condenser. Aspen data extracted for the analysis included feed/distillate/bottoms molar flow rates and mole fractions, feed K-values (for computing relative volatilities), the reflux flow $L$and distillate flow Dreported by RADFRAC, the feed thermal condition q, and tray-hydraulics quantities (internal vapor traffic and phase densities) from the Hydraulic Plots. Consistent with the Aspen split defined for

SAF and diesel-range material, the light key (LK) was defined as C16 and the heavy key (HK) as C18. The feed quality was taken as $q = 0.96$, indicating a predominantly liquid feed entering the column.

The minimum number of theoretical stages was estimated using the Fenske equation, which provides the minimum stage requirement under the ideal limiting case of total reflux:

$$N_{\min} = \frac{\ln\left[\left(\frac{x_{LK}}{x_{HK}}\right)_{dist} \Big/ \left(\frac{x_{LK}}{x_{HK}}\right)_{bot}\right]}{\ln\left(a_{LK/HK}\right)} \qquad \Rightarrow \qquad \boxed{N_{\min} = 4.0 \text{ theoretical stages}}$$

A small $N_{\min}$ indicates that the specified LK/HK cut is feasible from a volatility standpoint and does not require an exceptionally tall tower under ideal conditions.

To apply the multicomponent Underwood method, relative volatilities were computed using Aspen feed K-values referenced to the heavy key:

$$a_i = \frac{K_i}{K_{HK}}$$

Feed K-values were used as a representative basis because the ordering and relative behavior of Kacross the component slate remains consistent, even though absolute K-values vary by stage temperature and pressure.

The Underwood root equation was then solved for ϕ using the feed mole fractions z_i, the computed α_i, and $q = 0.96$:

$$\sum_i \frac{q\, z_i}{a_i - \phi} = 1 \qquad \Rightarrow \qquad \boxed{\phi = 1.9982}$$

The solution was confirmed to fall within the expected bracket for an HK-referenced volatility set, $1 < \phi < \alpha_{LK}$. The minimum vapor rate was computed using the second Underwood equation:

$$V_{\min} = \sum_i \frac{a_i D x_{D,i}}{a_i - \phi} \quad \Rightarrow \quad \boxed{V_{\min} = 661.7 \text{ lbmol/hr}}$$

Considering that a total condenser is present, the minimum reflux flow was obtained from:

$$L_{\min} = V_{\min} - D \quad \Rightarrow \quad \boxed{L_{\min} = 47.6 \text{ lbmol/hr}}$$

with the distillate flow taken from Aspen as $D = 613.57$ lbmol/hr, the corresponding minimum reflux ratio was:

$$\left(\frac{L}{D}\right)_{\min} = \frac{L_{\min}}{D} \quad \Rightarrow \quad \boxed{\left(\frac{L}{D}\right)_{\min} = 0.0776}$$

The operating reflux ratio was taken from Aspen RADFRAC results:

$$\frac{L}{D} = \frac{49.35}{613.57} \quad \Rightarrow \quad \boxed{\frac{L}{D} = 0.0804}$$

The operating reflux ratio being only slightly above $(L/D)_{\min}$ indicates operation near minimum reflux, which is consistent with requiring a larger number of stages than the minimum-stage limit.

The Gilliland correlation was then used to estimate the actual number of theoretical stages at the chosen operating reflux ratio. The Gilliland X parameter was computed as:

$$X = \frac{\left(\frac{L}{D}\right) - \left(\frac{L}{D}\right)_{\min}}{\left(\frac{L}{D}\right) + 1} \quad \Rightarrow \quad \boxed{X = 0.00265}$$

Using the correlation form for Y:

$$Y = \frac{1 - X^{0.0031}}{1 - 0.99357\, X^{0.0031}} \quad \Rightarrow \quad \boxed{Y = 0.7428}$$

The stage requirement was then obtained from the standard Gilliland relationship:

$$Y = \frac{N - N_{\min}}{N + 1} \quad \Rightarrow \quad N = \frac{Y + N_{\min}}{1 - Y} \quad \Rightarrow \quad \boxed{N = 18.48 \text{ stages}}$$

To convert theoretical stages to actual trays, an overall tray efficiency of $E = 0.80$ was applied:

$$N_{act} = \frac{N}{E} \quad \Rightarrow \quad \boxed{N_{act} = 23.10 \approx 23 \text{ trays}}$$

This tray count is consistent with a practical vacuum fractionation cut and reflects the expected increase in required stages when operating close to minimum reflux.

Column height was estimated using a standard tray spacing of 2 ft and additional allowances for top and bottom equipment/disengagement regions:

$$H = (N_{act} - 1)(2\,ft) + 4\,ft\,(top) + 8\,ft\,(bottom) \quad \Rightarrow \quad \boxed{H = 56\ ft}$$

Finally, column diameter was estimated by tray hydraulics using the maximum internal vapor-traffic stage identified in Aspen Hydraulic Plots (Stage 21). Flooding velocity was estimated using the Souders–Brown form with tray capacity constant C, and a design velocity at 80% of flooding was applied:

$$u_f = C\sqrt{\frac{\rho_L - \rho_V}{\rho_V}} \quad \Rightarrow \quad \boxed{u_f = 3.34\ m/s}$$

$$u_{des} = 0.8u_f \quad \Rightarrow \quad \boxed{u_{des} = 2.67\ m/s}$$

Diameter was then computed using the vapor mass flow rate $\dot{m}_V$, vapor density ρ_V, and an active area fraction $\phi = 0.8$to account for tray downcomers and non-active cross-sectional area. A tray capacity constant of $C = 0.11$ m/s was used:

$$D = \sqrt{\frac{4\dot{m}_V}{\pi \rho_V \phi u_{des}}} \quad (\phi = 0.8,\ C = 0.11\ m/s) \quad \Rightarrow \quad \boxed{D = 4.16\ m = 13.6\ ft}$$

The resulting diameter reflects the large cross-sectional area typically required for vacuum towers, where low vapor density increases volumetric flow and hydraulics become controlling. Differences between the hand diameter and Aspen-reported diameter are expected because Aspen tray sizing incorporates additional constraints (e.g., downcomer backup, entrainment, weeping, and tray geometry assumptions) and may be governed by a different internal location depending on the controlling hydraulic limit.

For comparison to Aspen's reported stage count, RADFRAC reports theoretical stages only (excluding condenser and reboiler). Aspen reported 22 theoretical stages for C-401; applying the same overall tray efficiency assumption ($E = 0.80$) gives an estimated physical tray count of $22/0.80 \approx 27$trays, which is consistent in form with the hand-correction procedure used above. The Aspen results and hand calculations are compared in the table below

Figure 28: *Hand vs Aspen Calculation for Column C-401*

C-401	Hand Calculation	Aspen calculation
Column Trays	23 trays	27 trays

Column Diameter	13.6 ft	14.0 ft
Column Height	56 ft	70 ft

As shown in *Figure* , although Aspen predicts a larger column height than the hand estimate, the simulated sizing results remain consistent with typical industrial vacuum column dimensions and provide a reliable basis for preliminary sizing.

Appendix VII: Preliminary Utilities

The following, *Figure ,* displays the preliminary utilities needed to operate the process. These specifications were obtained from the process simulated on ASPEN v14.

***Figure 29**: Cooling Water Flowrates*	
Equipment	**Flowrate (gpm)**
E-101	68046
E-102	3339
E-201	64182
E-300	37355
E-401	7829
E-400A	23196
E-401A	2209
E-501	17975
E-500A	9589
Total	**233720**

Figure 30: *Electricity Requirement for Equipment*	
Equipment	**Power (kw)**
K-100	5.33
K-200	13204
K-201	11310
P-400	26.72
P-400A	0.023
P-401A	0.049
K-400	14.31
P-500	297
P-501	11
P-500A	0.17
K-500	2180
Total	**27048.6**

Figure 31*: Steam Requirements for Equipment*			
Equipment	**Duty (kw)**	**Flow (lbs/hr)**	**Type**
E-100	0.979	3.54	LP
E-400B	1011.3	3944.6	MP
DRY-100	73751.6	340447.7	HP
E-500	1337.2	6172.6	HP
E-500B	7621.0	35179.8	HP

E-401B	1540.5	7111.1	HP

Figure 32: *Total Steam Consumption by Type*

Type	Duty (kw)	Flow (lbs/hr)
LP steam	0.979	3.54
MP steam	1011.3	3.944.6
HP steam	84250.3	392859.34

Figure 33: *Natural Gas Consumption by Equipment*

Equipment	Duty (kw)	Flow (lbs/hr)
R-100	21736.06	107819.5
E-400	1857.551	9214.197
R-101	34166.13	169477.7
Total	**57759.74**	**286511.397**

Appendix VIII: Economic Evaluation

This appendix contains the detailed calculations supporting the economic analysis presented in Section 7. While the main report summarizes the key economic indicators and conclusions, the tables included here provide the full intermediate calculations used to determine the net present value (NPV), discounted cash flow rate of return (DCFROR), depreciation, taxable income, taxes, and discounted cash flows for both the MACRS-5 and MACRS-7 depreciation schedules. The appendix also includes the installed cost estimates for all major process equipment. Additionally, the complete datasets used for the sensitivity analysis are provided, showing how variations in capital cost, operating costs, and product revenue influence the final NPV. These detailed calculations are included for completeness and transparency, allowing the economic methodology and results presented in the main report to be fully verified.

years	Pre-tax Cash Flow	Depreciation	Taxable Income	Tax	CF	Fd	PV of CF	NPV
1	-245.1		-245.1	0.0	-245.1	0.9	-213.2	-213.2
2	-735.4		-735.4	0.0	-735.4	0.8	-556.1	-769.3
3	207.2	133.3	74.0	0.0	207.2	0.7	136.3	-633.0
4	458.8	228.4	230.4	19.6	439.2	0.6	251.1	-381.9
5	458.8	163.1	295.7	61.1	397.7	0.5	197.7	-184.2
6	458.8	116.5	342.3	78.4	380.4	0.4	164.5	-19.7
7	458.8	83.3	375.5	90.7	368.1	0.4	138.4	118.7
8	458.8	83.2	375.6	99.5	359.3	0.3	117.4	236.1
9	458.8	83.3	375.5	99.5	359.2	0.3	102.1	338.2
10	458.8	41.6	417.2	99.5	359.3	0.2	88.8	427.0
11	458.8	0.0	458.8	110.6	348.2	0.2	74.8	501.9
12	458.8	0.0	458.8	121.6	337.2	0.2	63.0	564.9
13	458.8	0.0	458.8	121.6	337.2	0.2	54.8	619.7
14	458.8	0.0	458.8	121.6	337.2	0.1	47.7	667.4
15	458.8	0.0	458.8	121.6	337.2	0.1	41.4	708.8
16	458.8	0.0	458.8	121.6	337.2	0.1	36.0	744.8
17	458.8	0.0	458.8	121.6	337.2	0.1	31.3	776.2
18	458.8	0.0	458.8	121.6	337.2	0.1	27.2	803.4
19	458.8	0.0	458.8	121.6	337.2	0.1	23.7	827.1
20	458.8	0.0	458.8	121.6	337.2	0.1	20.6	847.7
21	458.8	0.0	458.8	121.6	337.2	0.1	17.9	865.6
22	458.8	0.0	458.8	121.6	337.2	0.0	15.6	**881.2**

***Figure** 34**:** NPV analysis for MACRS-7 year*

years	Pre-tax Cash Flow	Depreciation	Taxable Income	Tax	CF	Fd	PV of CF	NPV
1	-245.1		-245.1	0.0	-245.1	0.9	-213.2	-213.2
2	-735.4		-735.4	0.0	-735.4	0.8	-556.1	-769.3
3	207.2	186.5	20.7	0.0	207.2	0.7	136.3	-633.0
4	458.8	298.4	160.4	5.5	453.3	0.6	259.2	-373.8
5	458.8	179.0	279.7	42.5	416.3	0.5	207.0	-166.9
6	458.8	107.4	351.3	74.1	384.6	0.4	166.3	-0.6
7	458.8	107.4	351.3	93.1	365.7	0.4	137.5	136.9
8	458.8	53.7	405.1	93.1	365.7	0.3	119.5	256.4
9	458.8	0.0	458.8	107.3	351.4	0.3	99.9	356.3
10	458.8	0.0	458.8	121.6	337.2	0.2	83.4	439.7
11	458.8	0.0	458.8	121.6	337.2	0.2	72.5	512.2
12	458.8	0.0	458.8	121.6	337.2	0.2	63.0	575.2
13	458.8	0.0	458.8	121.6	337.2	0.2	54.8	630.0
14	458.8	0.0	458.8	121.6	337.2	0.1	47.7	677.6
15	458.8	0.0	458.8	121.6	337.2	0.1	41.4	719.1
16	458.8	0.0	458.8	121.6	337.2	0.1	36.0	755.1
17	458.8	0.0	458.8	121.6	337.2	0.1	31.3	786.4
18	458.8	0.0	458.8	121.6	337.2	0.1	27.2	813.7
19	458.8	0.0	458.8	121.6	337.2	0.1	23.7	837.4
20	458.8	0.0	458.8	121.6	337.2	0.1	20.6	858.0
21	458.8	0.0	458.8	121.6	337.2	0.1	17.9	875.9
22	458.8	0.0	458.8	121.6	337.2	0.0	15.6	**891.5**

***Figure** 35: NPV analysis for MACRS-5 year*

7-year	net change	ISBL	VCOP	FCOP	Revenue
	-30%	1818	2001	1783	952
	-20%	1802	1925	1780	1225
	-10%	1787	1848	1776	1499
	0%	1772	1772	1772	1772
	10%	1756	1695	1768	2045
	20%	1741	1619	1764	2318
	30%	1726	1542	1760	2591

5-year	net change	ISBL	VCOP	FCOP	Revenue
	-30%	1823	2005	1789	958
	-20%	1808	1929	1785	1231
	-10%	1793	1853	1781	1504
	0%	1777	1777	1777	1777
	10%	1762	1701	1773	2050
	20%	1746	1625	1769	2323
	30%	1731	1549	1766	2597

***Figure** 36:Sensitivity Analysis Data for both MACRS years*

MACRS-7 year (%)	MACRS-5 year (%)
14.29	20
24.49	32
17.49	19.2
12.49	11.52
8.93	11.52
8.92	5.76
8.93	0
4.46	0

***Figure** 37: Depreciation Values*

Name	Installed Cost [USD]	MOC	F_m	New Cost [USD]
K-100	870,600	CS	1	870,600
E-100	59,200	CS	1	59,200
DRY-100	6,333,300	CS	1	6,333,300
R-100	16,000,000	CS	1	16,000,000
CYC-200	4,000,000	CS	1	4,000,000
E-200A	1,883,000	CS	1	1,883,000
E-200B	1,099,900	CS	1	1,099,900
E-200C	246,700	CS	1	246,700
V-200	261,400	CS	1	261,400
E-201	360,400	CS	1	360,400
V-201	270,500	CS	1	270,500
R-200	3,750,000	1.25CR-0.5MO Alloy	1.6	6,000,000
C-200	670,000	CS	1	670,000
E-300A	186,300	CS	1	186,300
E-300B	576,200	CS	1	576,200
K-300A	1,536,632	CS	1	1,536,632
K-300B	1,398,296	CS	1	1,398,296
K-300C	1,247,052	CS	1	1,247,052
R-300	10,000,000	316 S.S	1.3	13,000,000
E-301A	580,200	CS	1	580,200
E-301B	982,500	CS	1	982,500
E-301C	86,200	CS	1	86,200
V-300	229,900	CS	1	229,900
E-302	304,900	CS	1	304,900
V-301	151,800	CS	1	151,800
P-400	128,800	CS	1	128,800
E-400	350,000	CS	1	350,000
R-400	5,000,000	1.25CR-0.5MO Alloy	1.6	8,000,000
K-400	3,000,000	CS	1	3,000,000
E-401	199,200	CS	1	199,200
V-400	185,700	CS	1	185,700
C-400	3,000,000	CS	1	3,000,000
C-401	7,000,000	CS	1	7,000,000
P-500	293,500	CS	1	293,500
E-500	132,300	CS	1	132,300
K-500	7,000,000	CS	1	7,000,000
P-501	272,200	CS	1	272,200
R-500	12,500,000	2.25CR-0.5MO Alloy	1.9	23,750,000
E-501	219,400	CS	1	219,400
V-500	298,600	CS	1	298,600

***Figure** 38: Installed Costs and MOC of all equipment*

CW

Equipment	duty (G/hrJ)	flow (lb/hr)	gpm	Cost ($/hr)
E-100	0.0	40.3	0.1	0.0003
E-301C	4.2	221125.9	441.9	1.5
E-302	35.6	1881933.9	3760.9	12.6
E-501	16.8	888526.8	1775.6	6.0
C-500	9.0	473981.4	947.2	3.2
R-500	12.4	656330.6	1311.6	4.4
E-401	7.4	393189.2	785.7	2.6
C-400	2.1	110946.8	221.7	0.7
C-401	22.5	1188088.8	2374.3	8.0
E-200C	37.2	1965922.2	3928.7	13.2
E-201	3.3	172074.2	343.9	1.2
total	150.5	7952160.1	15891.6	53.3
			$/oper yr	426,302

Natural Gas

Equipment	duty (GJ//hr)	flow (lb/hr)	Cost ($/hr)
E-400	6.7	9229.5	40.2
R-200	121.8	167819.6	730.8
Total	128.5	177049.1	771.0
		$/oper yr	6,167,667

HP steam

Equipment	duty (G/hrJ)	flow (lb/hr)	Cost ($/hr)
C-401	5.51	7068.04	54.19
C-500	27.05	34684.72	265.90
E-500	4.63	5935.21	45.50
C-400	3.65	4680.55	35.88
Total	40.84	52368.51	401.47
		$/oper yr	3,211,723

LP Steam Gen

Equipment	duty (G/hrJ)	flow (lb/hr)	Cost ($/hr)
E-301B	35.1	35337.4	273.3
E-200B	38.0	38250.2	295.9
E-300A	3.1	3086.0	23.9
E-300B	32.0	32224.3	249.3
Total	108.3	108898.0	842.3
		$/oper yr	6,738,630

Electricity

Equipment	duty (kWh)	Cost ($/hr)
K-100	3.9	0.2
K-300A	6589.0	369.6
K-300B	5617.5	315.1
K-300C	4627.5	259.6
P-500	220.0	12.3
P-501	8.2	0.5
K-500	1239.3	69.5
P-400	26.8	1.5
K-400	14.3	0.8
Total	18346.5	1029.2
	$/oper yr	8,233,899

HP Steam Gen

Equipment	duty (G/hrJ)	flow (lb/hr)	Cost ($/hr)
E-301A	21.0	26979.8	206.3
E-200A	561.6	722014.4	5520.3
Total	582.6	748994.1	5726.6
		$/oper yr	45,812,527

***Figure** 39**:** Utility Consumption for each equipment*

Appendix IX: Bibliography

1. ASTM International. *Standard Specification for Aviation Turbine Fuels*; ASTM International: West Conshohocken, PA, 2023.
2. ASTM International. *Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons*; ASTM D7566-23; ASTM International: West Conshohocken, PA, 2023.
3. Béalu, Z.; Burger, J.; Friedemann, M. *Process Simulation and Optimization of Alternative Liquid Fuels Production: A Techno-Economic Assessment of the Production of HEFA Jet Fuel*; Master's Thesis, German Aerospace Center (DLR), 2017. Available online: https://elib.dlr.de/117806/1/Masterarbeit_Zoe_Bealu.pdf (accessed 2025).
4. Boymans, E.; Ganjkhanlou, Y.; Agirrezabal-Telleria, I.; Viar, N.; Vreugdenhil, B. Syngas to Sustainable Aviation Fuel: Emerging Catalysts and Routes. *Appl. Catal. A* 2025, *708*, 120554. https://doi.org/10.1016/j.apcata.2025.120554.
5. Consonni, S.; Larson, E. Biomass-Gasifier/Aeroderivative-GT Combined Cycles: Part A—Performance Analysis. *J. Eng. Gas Turbines Power* 2003, *125* (3), 745–752.
6. Curcio, E. Techno-Economic Analysis of Hydrogen Production: Costs, Policies, and Scalability in the Transition to Net-Zero. *Int. J. Hydrogen Energy* 2025, *128* (15), 473–487. https://doi.org/10.1016/j.ijhydene.2025.04.013.
7. Dry, M. E. The Fischer–Tropsch Process: 1950–2000. *Catal. Today* 2002, *71* (3–4), 227–241.

8. Fortunate Business Insights. *Sustainable Aviation Fuel (SAF) Market Size, Share & COVID-19 Impact Analysis.* Available online: https://www.fortunebusinessinsights.com/sustainable-aviation-fuel-saf-market-111560 (accessed 2025).
9. Geleynse, S.; Brandt, K.; Garcia-Perez, M.; Wolcott, M.; Zhang, X. The Alcohol-to-Jet Conversion Pathway for Drop-In Biofuels: Techno-Economic Evaluation. *ChemSusChem* 2018, *11* (21), 3728–3741. https://doi.org/10.1002/cssc.201801690.
10. Holcombe, T. *The New CatFT® Process.* Green Impact Fuels, LLC, 2014. SlideShare. https://www.slideshare.net/slideshow/catftr-fischer-tropsch-process-presentation/49036312 (accessed 2025).
11. Idaho National Laboratory. *Production of Fischer–Tropsch Synfuels at Nuclear Plants*; INL/RPT-22-69047, 2022. Available online: https://inldigitallibrary.inl.gov/sites/sti/sti/Sort_63673.pdf.
12. International Civil Aviation Organization (ICAO). *SAF Production Facilities.* https://www.icao.int/SAF/SAF-production-facilities (accessed 2025).
13. Johnson Matthey. DG Fuels Selects Johnson Matthey and bp's FT CANS™ Technology for Its First Sustainable Aviation Fuel Plant. Available online: https://matthey.com/products-and-markets/energy/sustainable-aviation-fuel-saf-technology/ft-cans-technology (accessed 2025).
14. Light Water Reactor Sustainability Program. *Production of Fischer–Tropsch Synfuels at Nuclear Plants*; Idaho National Laboratory, 2022.

15. Meurer, A.; Kern, J. Fischer–Tropsch Synthesis as the Key for Decentralized Sustainable Kerosene Production. *Energies* 2021, *14* (7), 1836. https://doi.org/10.3390/en14071836.
16. Rase, H. F. *Handbook of Commercial Catalysts: Heterogeneous Catalysts*; CRC Press: Boca Raton, FL, 2000.
17. Speight, J. G. *The Chemistry and Technology of Petroleum,* 5th ed.; CRC Press: Boca Raton, FL, 2014.
18. Trading Economics. Cobalt Price Data & Historical Charts. https://tradingeconomics.com/commodity/cobalt (accessed 2025).
19. Trading Economics. Naphtha Historical Price Data. https://tradingeconomics.com/commodity/naphtha (accessed 2025).
20. U.S. Department of Energy. *Sustainable Aviation Fuel Grand Challenge.* https://www.energy.gov/eere/bioenergy/sustainable-aviation-fuel-grand-challenge (accessed 2025).
21. U.S. Energy Information Administration (EIA). Gasoline and Diesel Fuel Update. https://www.eia.gov/petroleum/gasdiesel/ (accessed 2025).
22. U.S. Energy Information Administration (EIA). U.S. Gulf Coast Kerosene-Type Jet Fuel Spot Price. https://www.eia.gov/dnav/pet/hist/eer_epjk_pf4_rgc_dpgD.htm (accessed 2025).
23. van der Laan, G. P.; Beenackers, A. A. C. M. *Kinetics and Selectivity of the Fischer–Tropsch Synthesis.* Appl. Catal. A 1999, *193*, 39–53
24. U.S. Energy Information Administration (EIA). *Louisiana: Wood and Biomass Waste Price and Expenditure Estimates (Table F46).* State Energy Data System (SEDS), 2024. https://www.eia.gov/state/seds/ (accessed 2025)

25. Wankat, P. C. *Separation Process Engineering: Includes Mass Transfer Analysis*. 4th ed. Pearson, 2022.

26. King, C. J. *Separation Processes*. 2nd ed. McGraw–Hill, 1980.

27. McCabe, W. L.; Smith, J. C.; Harriott, P. *Unit Operations of Chemical Engineering*. 7th ed. McGraw–Hill, 2005.

28. NPTEL. "Module #7 (Tray Columns / Hydraulics): Capacity parameter (Souders–Brown-type), flow parameter, and design velocity as a fraction of flooding." NPTEL course notes (PDF), accessed 2026.

29. AccessEngineering (Perry's Handbook platform). "Equipment for Distillation and Gas Absorption (tray and downcomer/active area guidance; downcomer region typically 10–20% of tray area)." AccessEngineering online reference, accessed 2026.

30. U.S. Energy Information Administration (EIA). *Energy Costs.* https://www.eia.gov/electricity/annual/html/epa_02_10.html (accessed 2026).

31. U.S. Energy Information Administration (EIA). *Today in Energy: Daily Energy Prices.* https://www.eia.gov/todayinenergy/prices.php (accessed 2026).

32. Business Analytiq. *Oxygen Prices 2026 | February Price Chart, Trend & Forecast.* https://businessanalytiq.com/procurementanalytics/index/oxygen-price-index/ (accessed 2026).

33. Business Analytiq. *Green Hydrogen Price Index.* https://businessanalytiq.com/procurementanalytics/index/green-hydrogen-price-index/ (accessed 2026).

34. U.S. Congressional Research Service. *Sustainable Aviation Fuel Tax Credit (CFPC) Overview.* https://www.congress.gov/crs-product/IF12502 (accessed 2026).

35. ZipRecruiter. *Plant Operator Salary in Louisiana: Hourly Rate.* https://www.ziprecruiter.com/Salaries/Plant-Operator-Salary--in-Louisiana (accessed 2026).

36. Dry, M. E. *The Fischer–Tropsch Process: 1950–2000.* Catalysis Today, 2002.

37. Speight, J. G. *The Chemistry and Technology of Petroleum.* 5th ed. CRC Press, 2014.

38. Ancheyta, J. *Hydroprocessing of Heavy Oils and Residua.* CRC Press, 2007.

39. Gary, J. H.; Handwerk, G. E.; Kaiser, M. J. *Petroleum Refining: Technology and Economics.* 5th ed. CRC Press, 2007.

40. Prudhomme, J. *Upgrading – Hydrocracker/Hydrotreater – Run HTHC-PU-2011-2 – 3 Day Test Run.* Rentech, Inc., Technical Report, March 5, 2012.

41. Towler, G.; Sinnott, R. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design.* 3rd ed.; Butterworth-Heinemann: Oxford, UK, 2022.

42. 42. U.S. Department of Energy, National Energy Technology Laboratory (NETL). *Gasifipedia: Syngas Contaminant Removal and Conditioning.* Overview. https://www.netl.doe.gov/research/coal/energy-systems/gasification/gasifipedia/cleanup

43. Airlines for America. *Sustainable Aviation Fuel Price Comparison.* Overview. https://www.airlines.org/dataset/sustainable-aviation-fuel-price-comparison/

44. Aegis Hedging. (2025, December 8). *LCFS & RIN pricing report through December 5th, 2025*. https://www.aegis-hedging.com/insights/lcfs-rin-pricing-report-through-december-8th-2025

45. Intratec Solutions. (2026). *Naphtha price in the United States*. https://www.intratec.us/solutions/energy-prices-markets/commodity/naphtha-price-united-states

46. University of Michigan Center for Sustainable Systems. (2024). *Municipal solid waste factsheet*. https://css.umich.edu/publications/factsheets/material-resources/municipal-solid-waste-factsheet

47. MCF Environmental Services. (2022). *Hazardous waste disposal costs: What to know about transportation fees*. https://mcfenvironmental.com/hazardous-waste-disposal-costs-what-to-know-about-transportation-fees/